GY457

Applied Urban and Real Estate Economics

Topic 8:

Housing Issues in China: Facts, Causes, and Policies





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Housing Issues in China: Facts, Causes, and Policies

Introduction

- Xiaolun Yu
  - Lecturer of Real Estate Finance at Henley Business School, University of Reading
  - BSc (Tongji, Shanghai) → Exchange (St.Gallen, Switzerland) → MSc REEF (LSE) → PhD (LSE)
  - Research: Urban Economics, Real Estate, Policy Evaluation, Applied Microeconomics (with a focus on China and the UK)
  - Teaching: Urban Economics, Real Estate Economics, Real Estate Finance/Investment/Valuation, Spatial Econometrics, Housing Policies, etc.
  - Office hours will be available
  - Contact: xiaolun.yu@henley.reading.ac.uk

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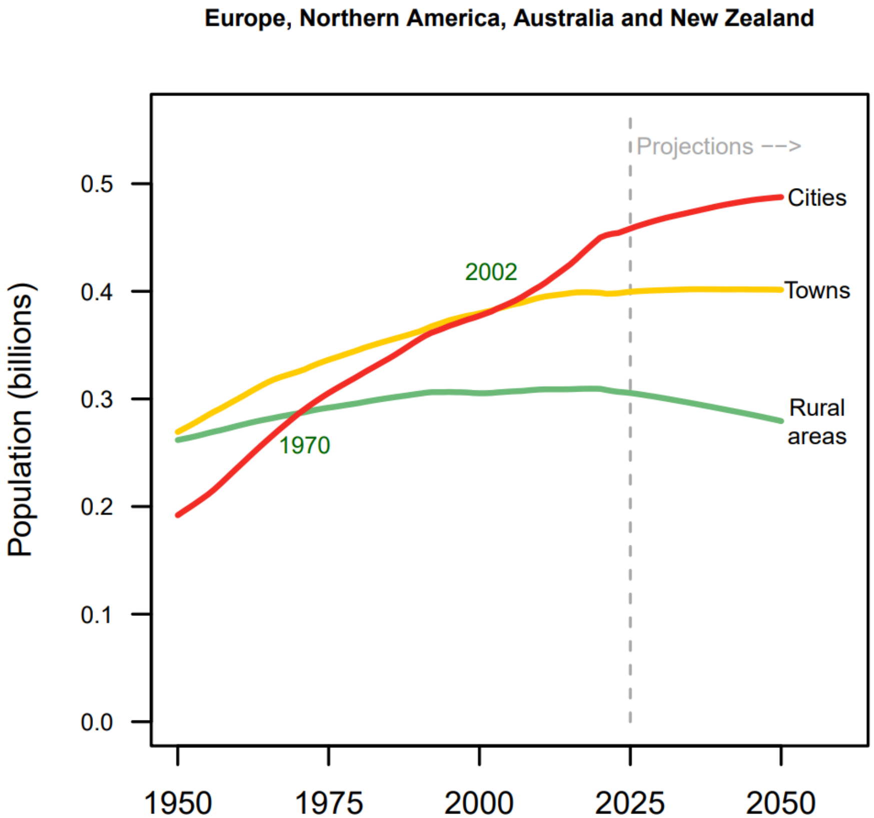
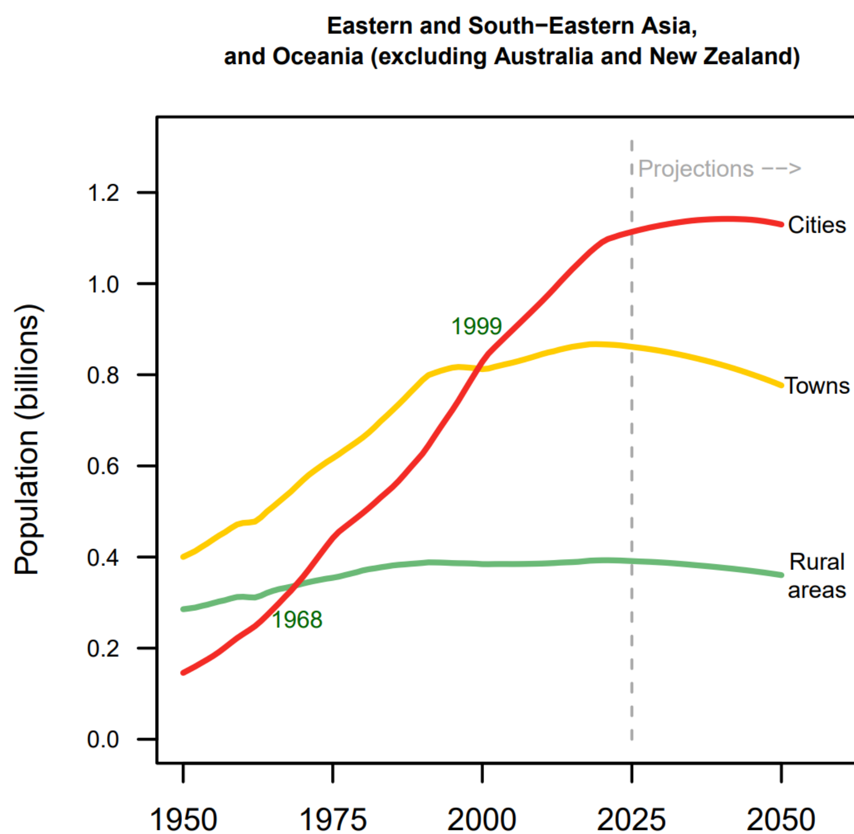
Stylized fact #1: Asia and particularly China have been urbanizing at a breath-taking pace



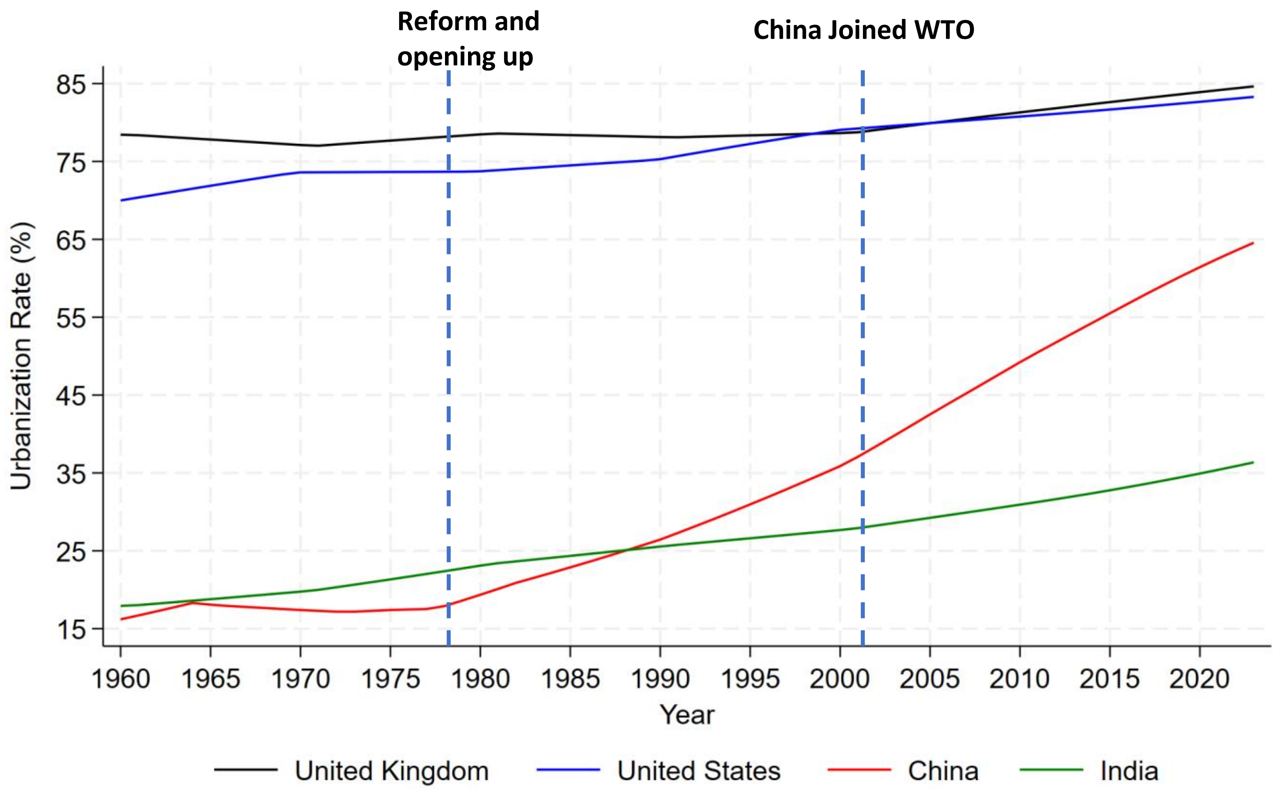


Note: “Cities”, “towns” and “rural areas” refer to standardized categories based on population size and density

Source: World Urbanization Prospects 2025 (United Nations, 2025)

Stylized fact #1: Asia and particularly China have been urbanizing at a breath-taking pace

- Urbanization rates (share population living in urbanized areas)

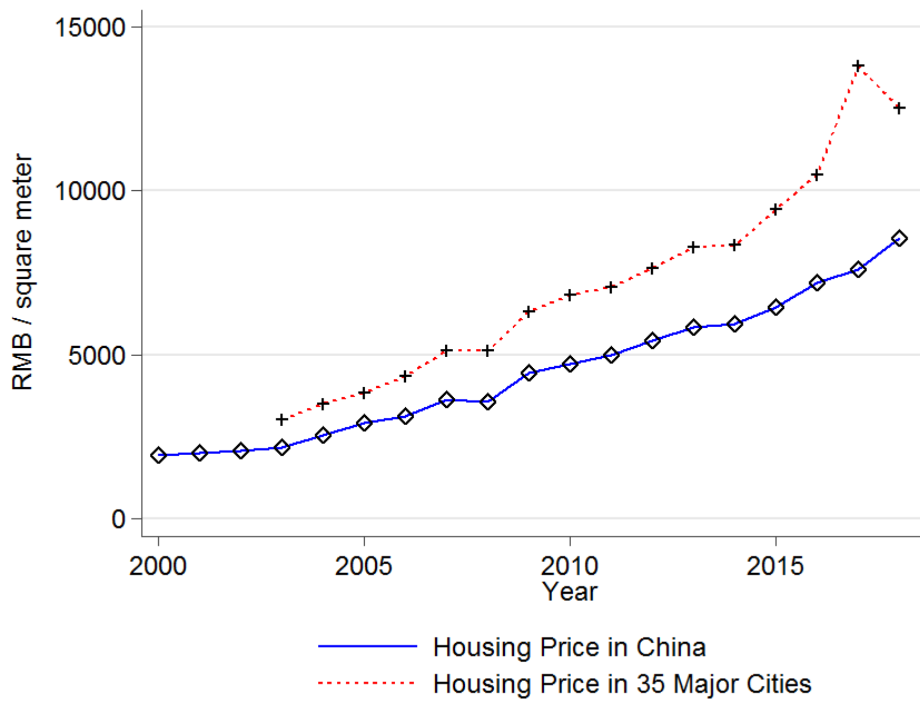


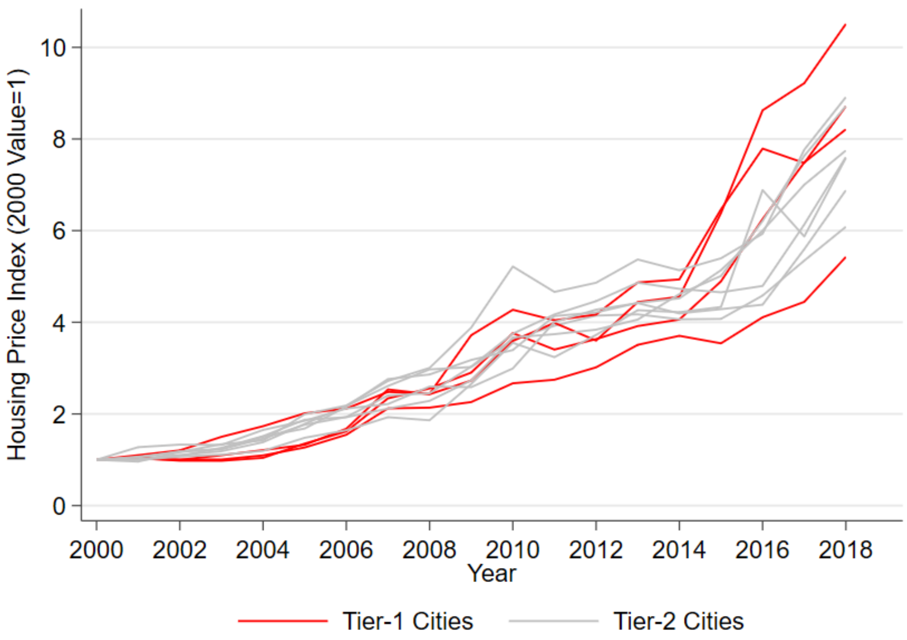
Source: World Bank

⇨ China has been urbanizing rapidly since at least 1950 (with exception of Cultural Revolution years) but now slowing down. The urbanization rate in China is 67% in 2024 (National Bureau of Statistics)

Stylized fact #2: Urbanization is associated with urban housing affordability crisis

- Before COVID: Avg. nominal HP increases by 313% (annual growth rate 8%) in 35 major cities in China from 2003 to 2018
  - Nominal HPs in Tier-1 cities increase even faster



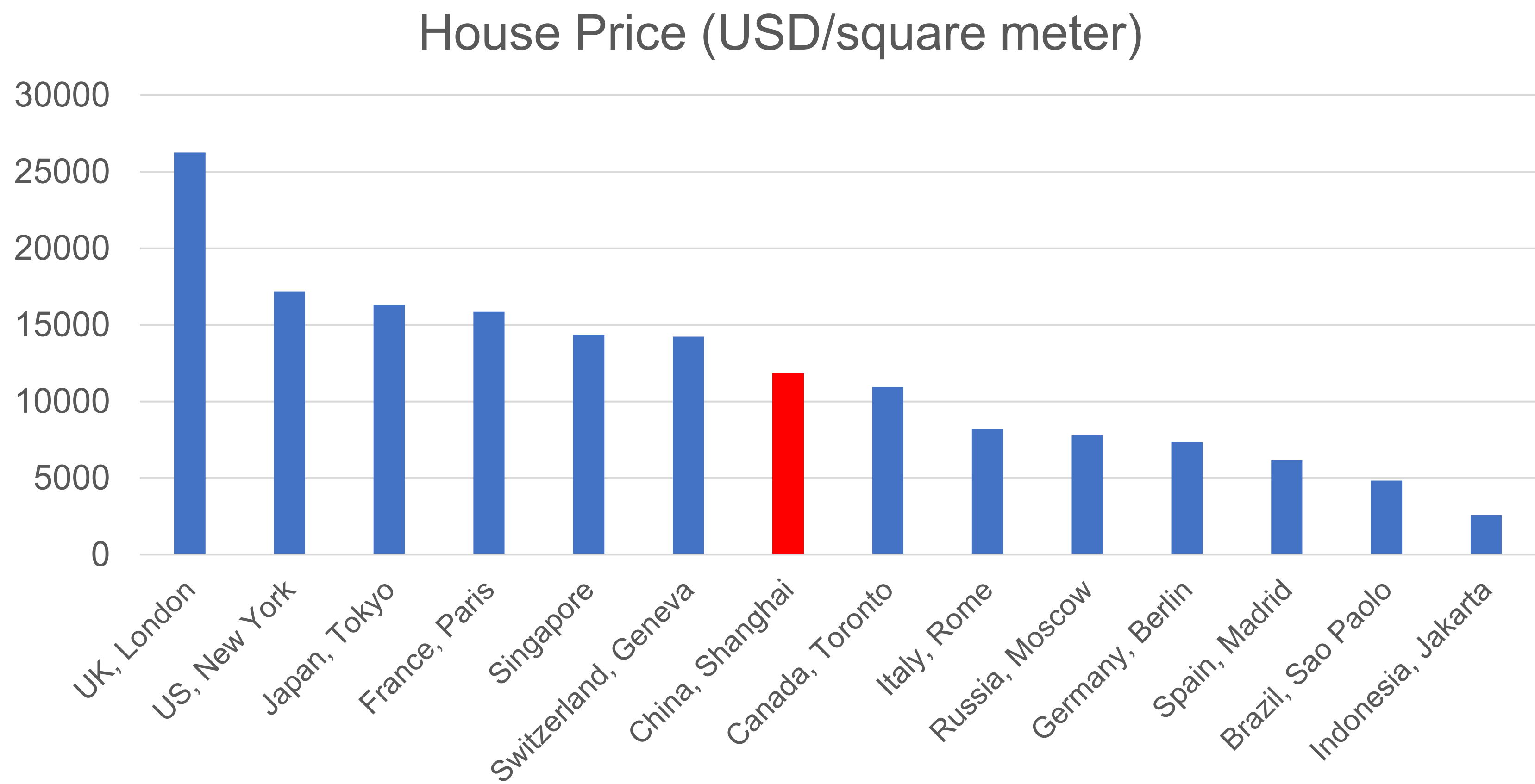


1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC)

Source: National Bureau of Statistics, city-level and province-level statistical yearbooks & self calculation

Stylized fact #2: Urbanization is associated with urban housing affordability crisis

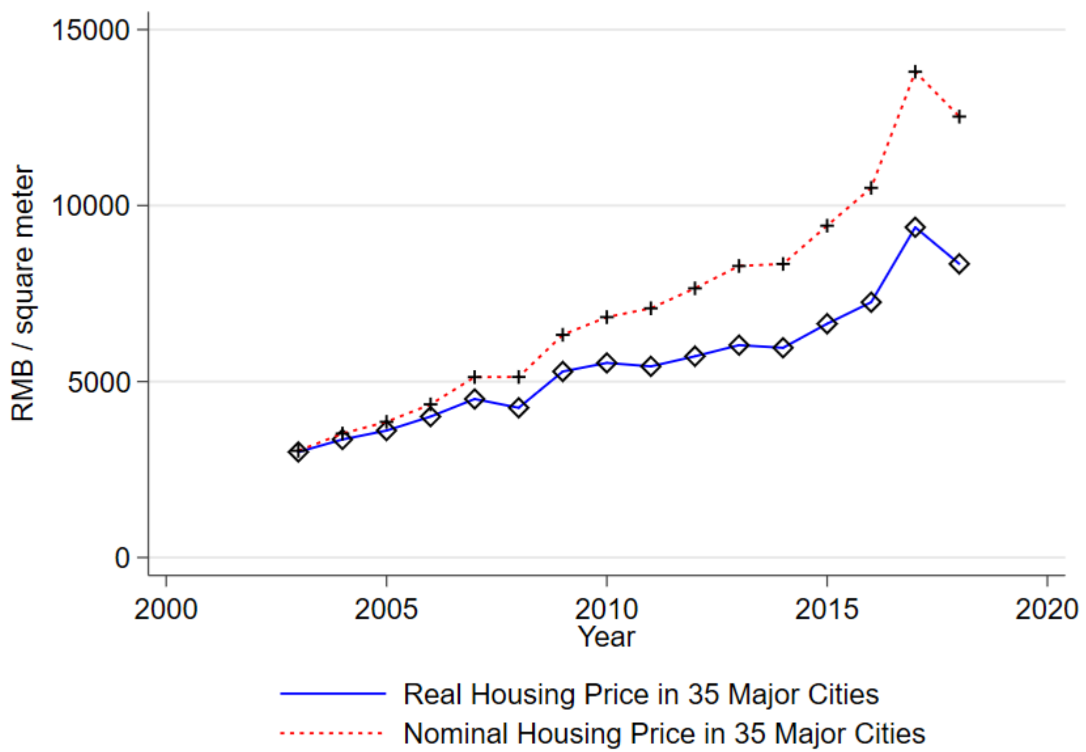
- HPs (in 2019 US$) of big cities in China are comparable with HPs in superstar cities in developed countries



Source: Global Property Guide
Last accessed: 10 Oct 2019

But what about *real* house prices?

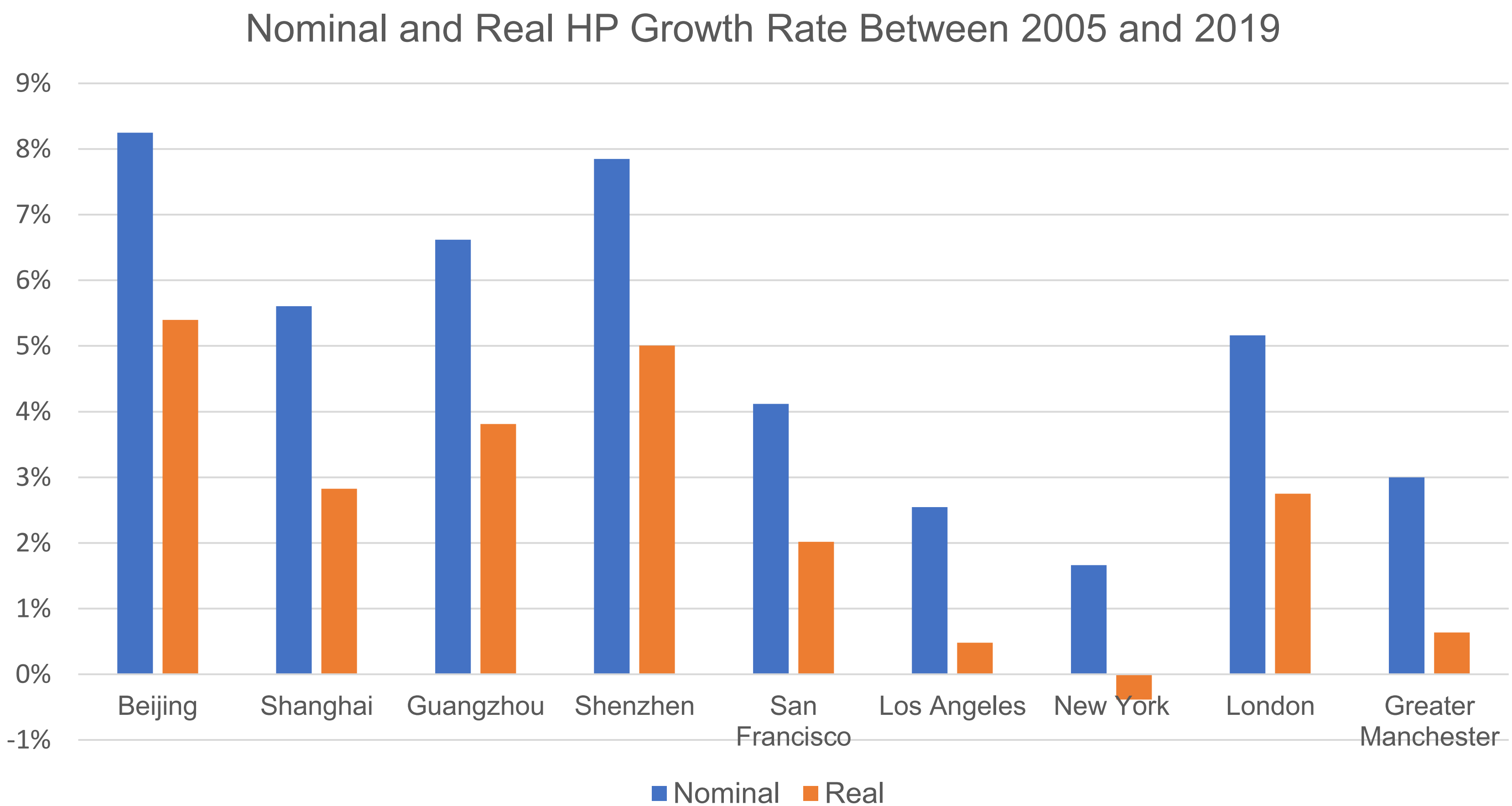
- Nominal HPs represent the monetary values of housing. Real HP is the nominal HP adjusted for inflation
- Avg. real HPs increase by 178% (ann. growth rate of 4% real vs. 8% nominal) in 35 major cities (2003-2018, 2003 RMB)



Source: National Bureau of Statistics & self calculation

Stylized fact #2: Urbanization is associated with urban housing affordability crisis

- Between 2005 and 2019, HP growth rate is relatively high in China compared with HP growth rate in the US and UK



Source: Land Registry, Federal Housing Finance Agency, National Bureau of Statistics & self calculation

Excursus: Chinese city tier system

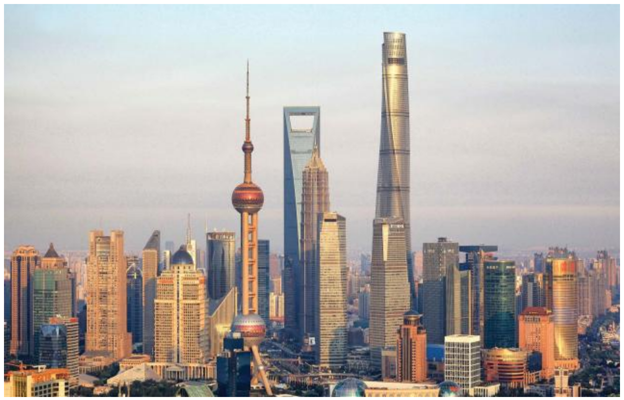
- The Chinese city tier system is a hierarchical classification of Chinese cities. There are no such official lists in China
- However, it is frequently referred to by various media publications. Cities in different tiers reflect differences in consumer behavior, income level, population size, business opportunity, etc.

Tier	Cities	Number of cities
Tier 1	Beijing, Shanghai, Guangzhou, Shenzhen	4
New Tier 1	Chengdu, Hangzhou, Wuhan, Chongqing, etc.	15
Tier 2	Xiamen, Fuzhou, Wuxi, Hefei, etc.	30
Tier 3	Weifang, Baoding, Zhenjiang, Yangzhou, etc.	70

Source: Yicai Global 2017

Chinese city tier system

- Tier-1 city: Shanghai
- Tier-2 city: Hefei
- Tier-3 city: Suqian



Population (Million)	24.2
Disposable Income (RMB)	57,692
GDP (Billion RMB)	2,818
GDP per capita (RMB)	116,562
House price (RMB/m^2)	25,910

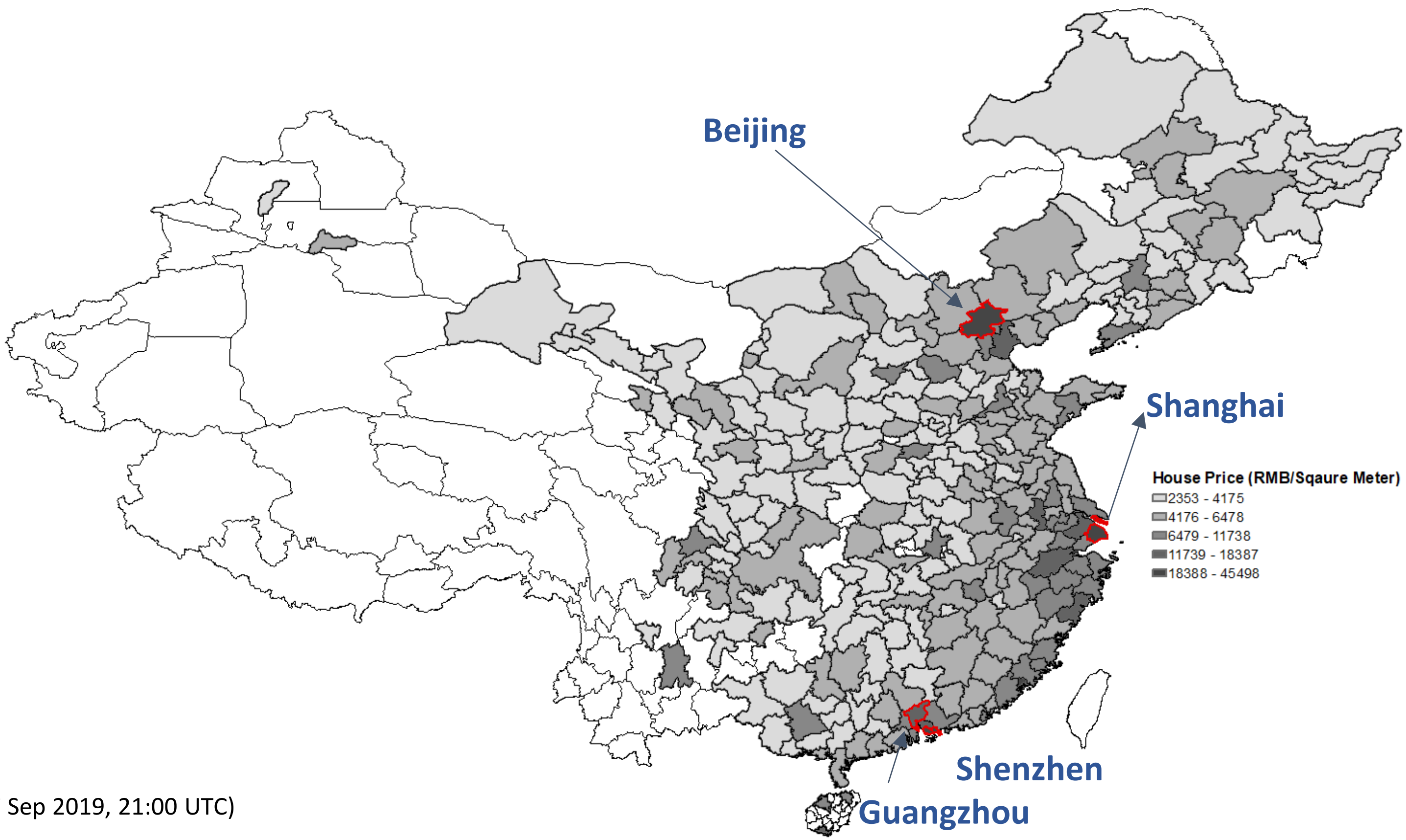
Population (Million)	7.9
Disposable Income (RMB)	34,852
GDP (Billion RMB)	627
GDP per capita (RMB)	80,138
House price (RMB/m^2)	9,312

Population (Million)	4.9
Disposable Income (RMB)	24,086
GDP (Billion RMB)	235
GDP per capita (RMB)	48,311
House price (RMB/m^2)	4,514

Note: data in 2016, 1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC).

Stylized fact #2: Urbanization is associated with urban housing affordability issues

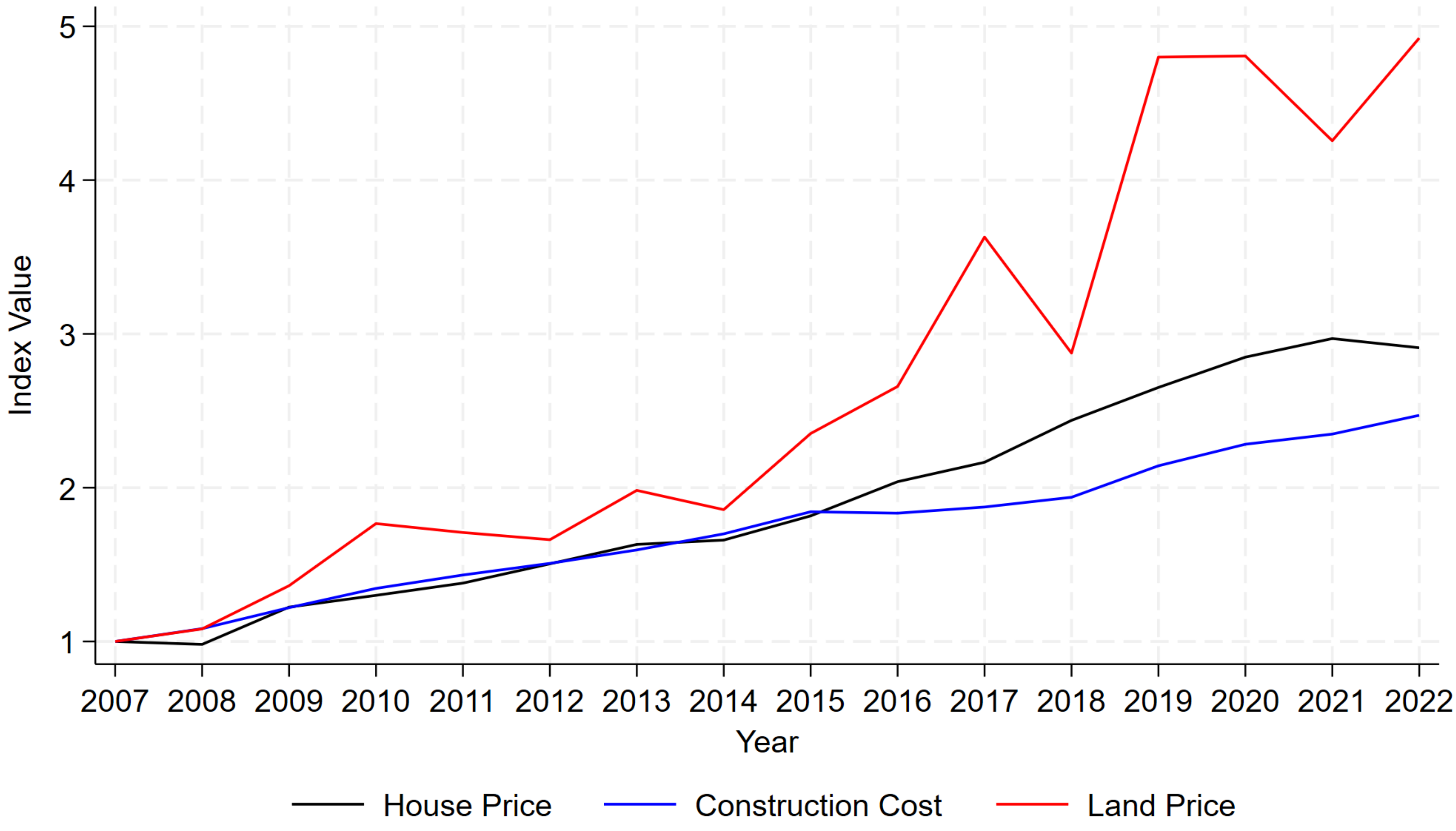
- In consistent with economic developments, HPs in Tier-1, Tier-2 cities and cities along the southeast coast are higher



Notes:
1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC)
HP data in 2016
A 100 square meter house in Beijing in 2018: 3,742,000 RMB (411,620 GBP)

Stylized fact #3: Strong HP growth is mainly driven by land prices and not by construction costs

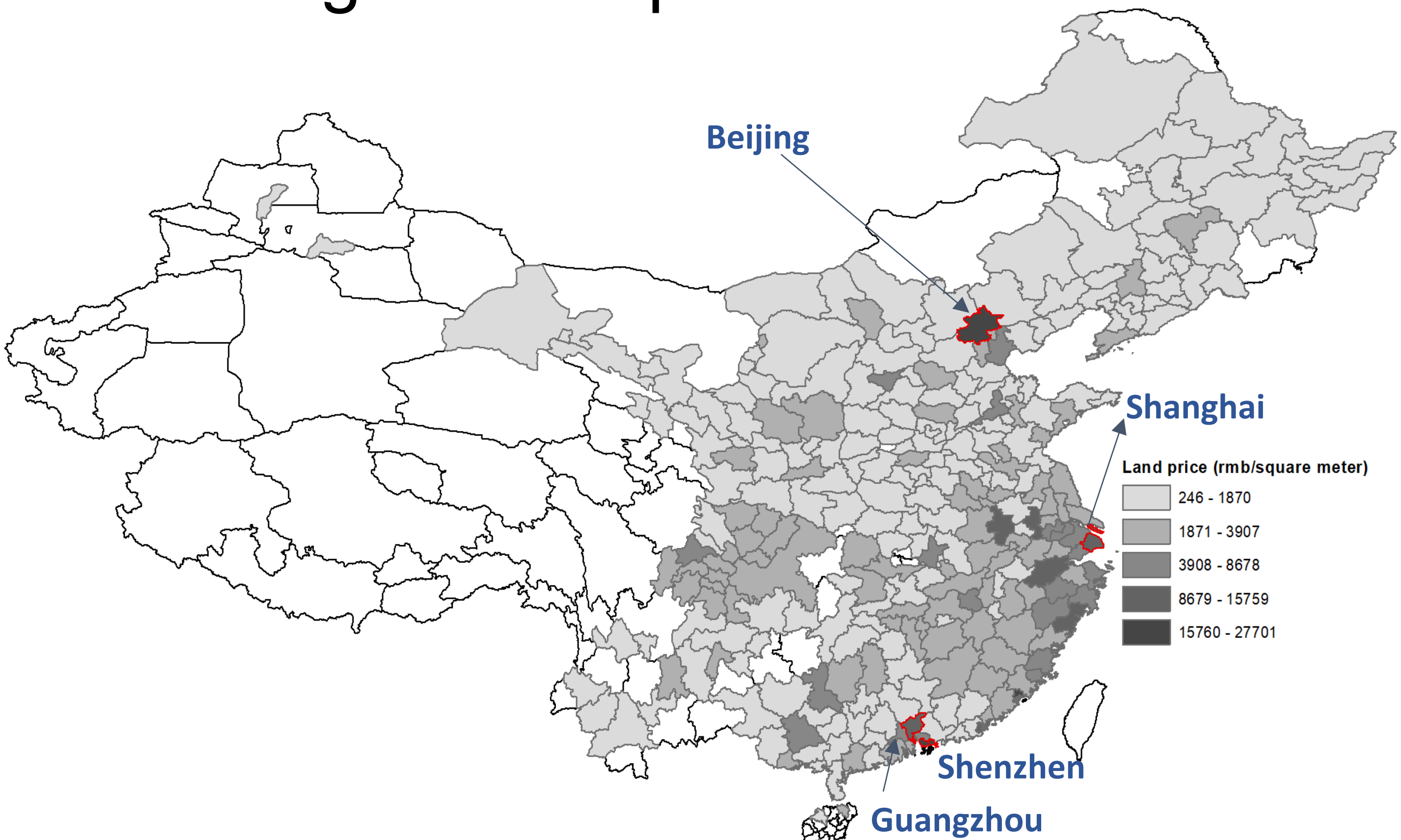
- Land prices increase faster than construction costs



Source: National Bureau of Statistics & self calculation

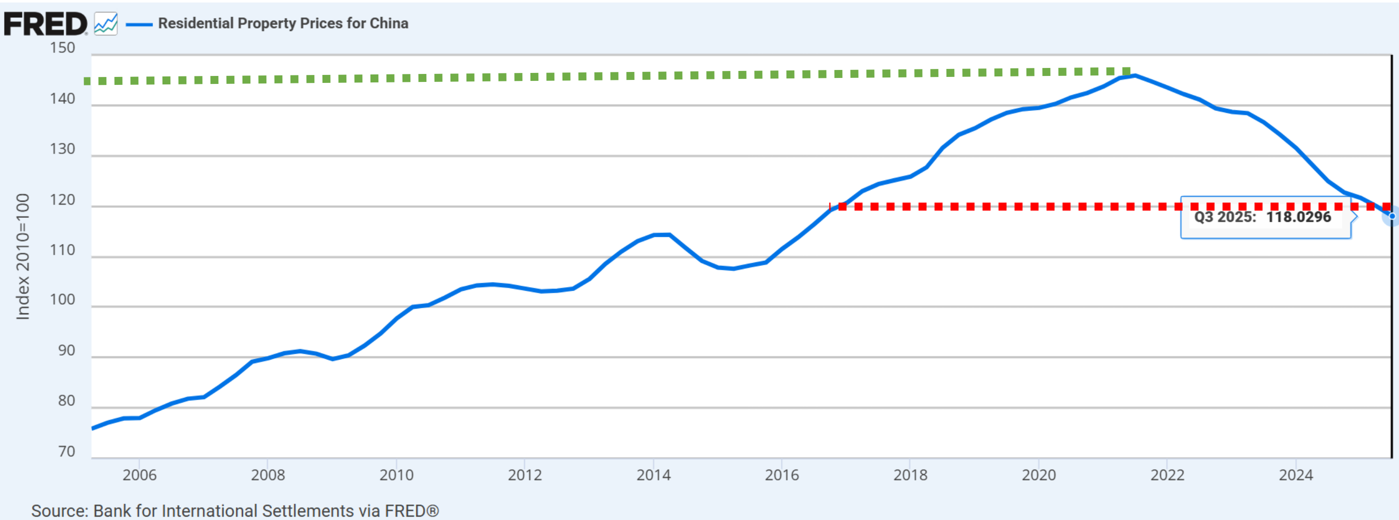
Stylized fact #3: Strong HP growth is mainly driven by land prices and not by construction costs

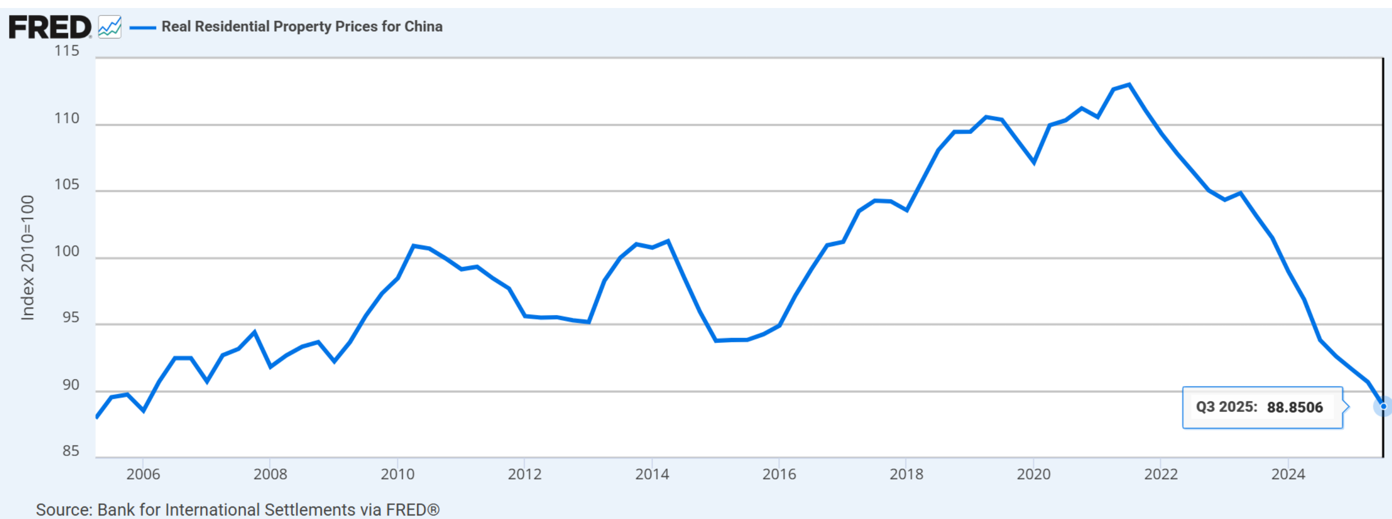
- Tier-1, Tier-2 cities, and cities along the southeast coast have higher land prices



Note: 1 RMB = 0.11 GBP (27 Sep 2019, 21:00 UTC)
Land price data in 2016

Stylized fact #4: HPs decline after COVID



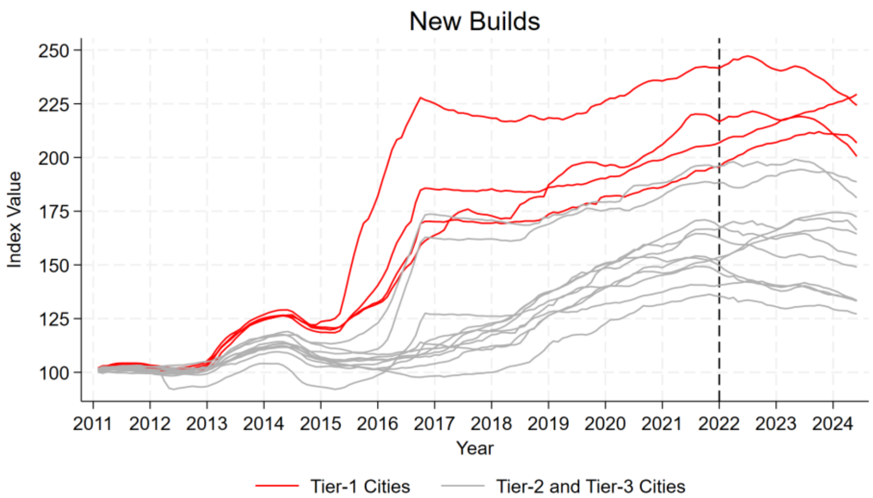


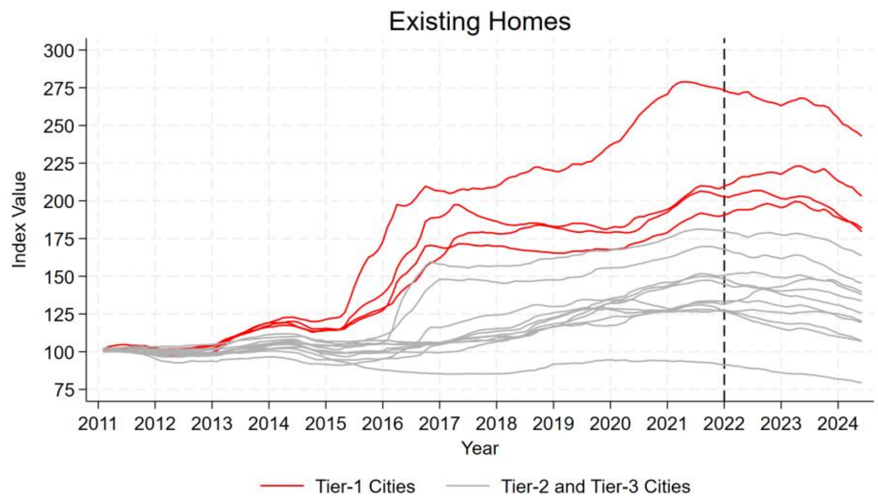
24% nominal HP decrease and 27% real HP decrease in Q3 2025 compared with Q3 2021

Sources: https://fred.stlouisfed.org/series/QCNN628BIS#
https://fred.stlouisfed.org/series/QCNR628BIS#
Coverage includes Q1 2007-Q4 2015: New dwellings in 70 cities in China; from Q1 2016: Existing buildings in 70 cities in China. The real HP series is deflated using CPI.

Stylized fact #4: Spatial variations in HPs after COVID

- Based on the 70 large and medium-sized cities HP index:
  - 0.5% HP **increase** (new builds) and 8.1% HP **decrease** (existing homes) in Tier 1 cities between September 2021 and May 2024
  - 6.6% HP **decrease** (new builds) and 12.3% HP **decrease** (existing homes) in lower Tier cities between September 2021 and May 2024



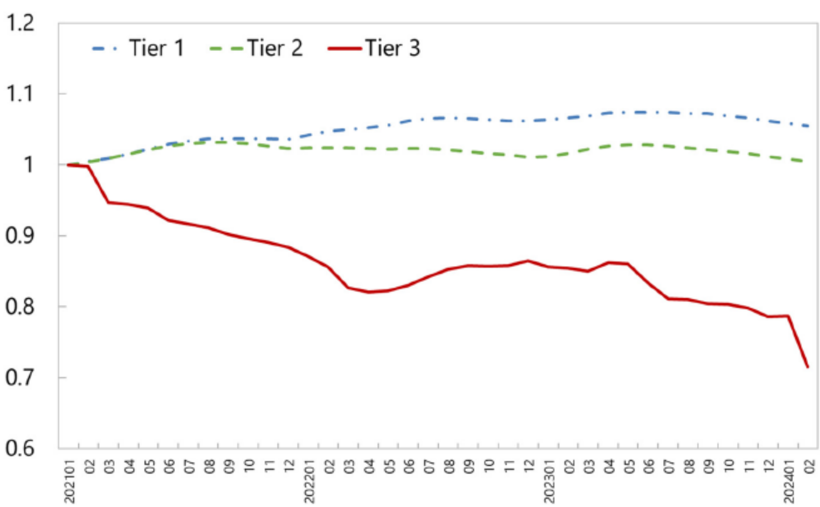


Source: National Bureau of Statistics & self calculation

Stylized fact #4: Spatial variations in HPs after COVID

- House prices have levelled off in Tiers 1 and 2 cities, but have fallen sharply in Tier-3 cities
- Real estate is running into diminishing returns to growth, mainly driven by Tier-3 cities

House Price Changes by City Tier (2021:01 to 2024:02)
(202101=1)



Sources: NBS and author calculations. See Appendix I.3.

Fig. 3. House Prices Change by City Tier (2021:01 to 2024:02).

Real Estate and Growth.

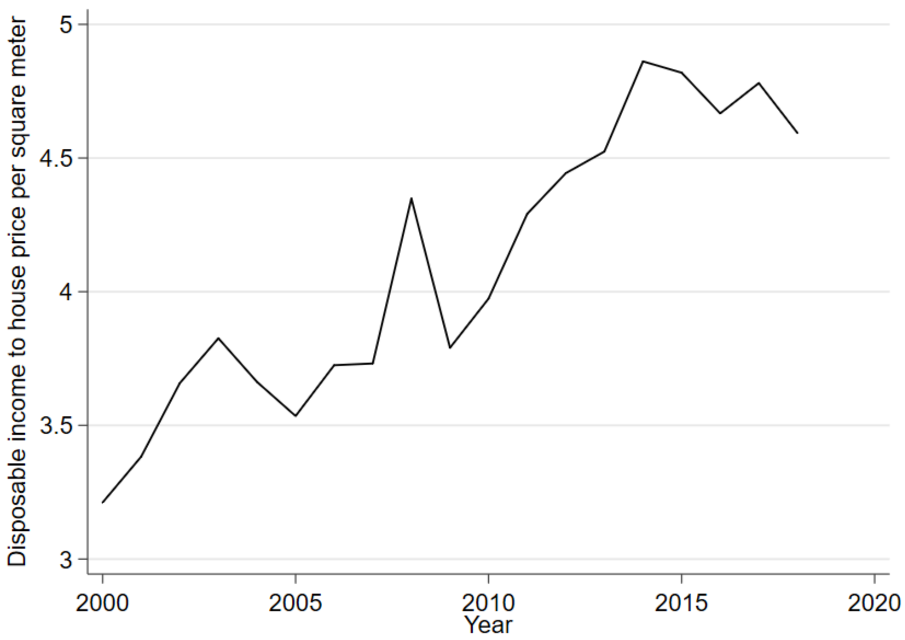
Variables	Real GDP growth
Instrumented real estate investment to GDP ratio	0.492***
	(0.064)
Instrumented real estate investment to GDP ratio × Period 09–15	−0.148**
	(0.068)
Instrumented real estate investment to GDP ratio × Period 16–19	0.045
	(0.082)
Instrumented real estate investment to GDP ratio × Period 20–22	−0.596***
	(0.085)
Real GDP per capita	−0.078***
	(0.007)
Population growth rate	0.014
	(0.035)
Urbanization rate	0.002***
	(0.000)
Constant	0.630***
	(0.054)
Number of observations	5689
R squared	0.341
Year fixed effect	YES
City fixed effect	YES

Source: Rogoff and Yang (2024)

Contents

Housing affordability

- Is Chinese housing boom driven by fundamentals? The ratio of disposable income relative to housing price per square meter is generally increasing over the past 20 years
- However, housing is very unaffordable in Tier-1 cities



Source: National Bureau of Statistics & self calculation

Housing affordability: international comparison

- Demographia uses *median housing price* relative to *median household income* to measure housing affordability and make international comparison

Nation	Affordable (3.0 & Under)	Moderately Unaffordable (3.1-4.0)	Seriously Unaffordable (4.1-5.0)	Severely Unaffordable (5.1 & Over)	Total	Median Market
Australia	1	1	5	16	23	5.7
Canada	12	16	5	17	50	4.0
China (Hong Kong)	0	0	0	1	1	20.9
Ireland	2	2	1	0	5	3.7
New Zealand	0	0	2	6	8	6.5
Singapore	0	0	1	0	1	4.6
United Kingdom	0	4	18	11	33	4.8
United States	56	70	34	28	188	3.5
TOTAL	71	93	66	79	309	4.0

Source: 15th Annual Demographia International Housing Affordability Survey: 2019

House price to income ratio in China

- Assumptions:
  - ➢ 2.62 person per household based on 2020 Census data
  - ➢ Average floor area per person = 36.52 m^2 based on 2020 Census data

$$Housing\ affordability = \frac{Housing\ Price\ per\ m^2 \times Floor\ Area\ Per\ Person\ \times Household\ Size}{Average\ Salary \times Household\ Size}$$

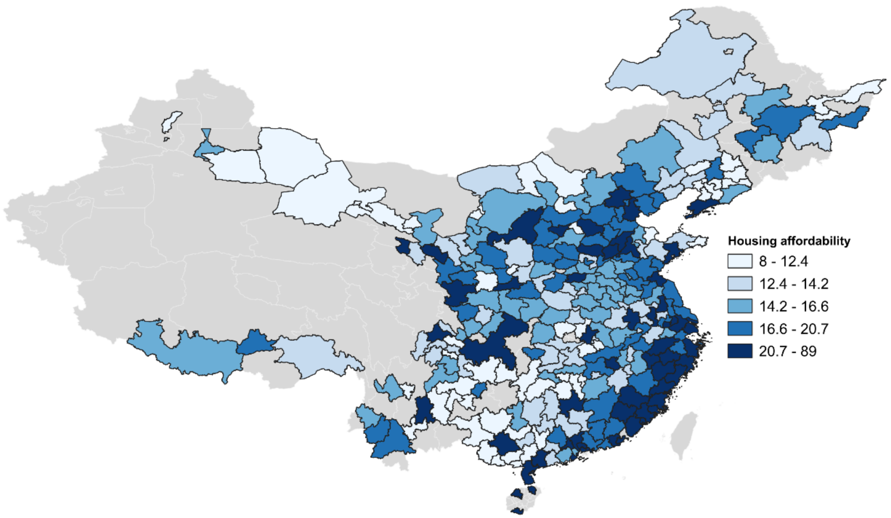
Tier-1 Cities	**Ratio**	**Tier-2 Cities**	**Ratio**
Beijing	21	Tianjin	11.6
Shanghai	18.2	Xiamen	17.1
Guangzhou	15	Nanjing	13.9
Shenzhen	31.8	Hangzhou	14.6
San Francisco	8.8	Wuhan	10.2
London	8.3	Chengdu	10.2

Source: 15th Annual Demographia International Housing Affordability Survey: 2019, National Bureau of Statistics, and China Statistical Yearbook. City level data in 2021 & self calculation. If household employment rate is not 100%, the ratio will be higher.

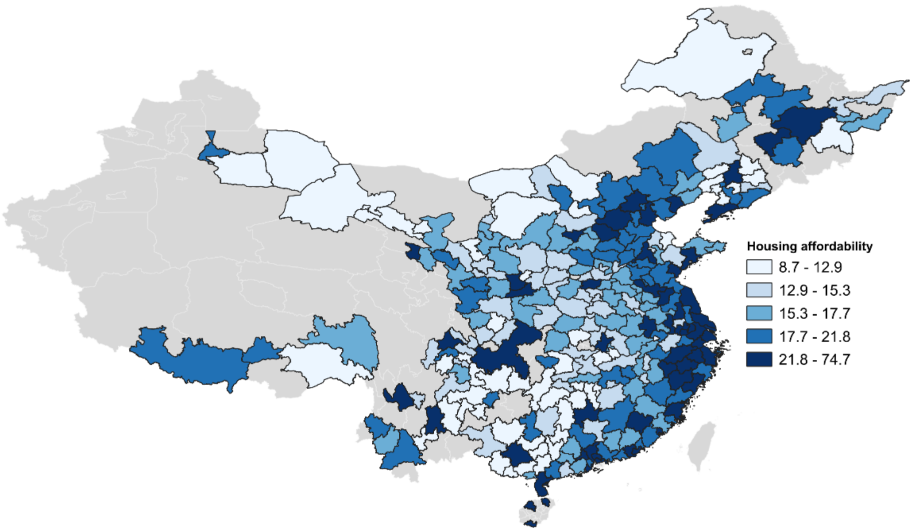
House price to income ratio in China — *cont.*

- Based on Demographia's criteria, housings are "severely unaffordable" in many Chinese cities
- There are also spatial variations in housing affordability status

Existing Homes



New Builds



Note: Both maps present the house price (defined as 100 square meters multiplied by the average house price per square) divided by the disposable income ratio in 2024

Source: Yu (2026)

Other measures of affordability in China

- Li, Qin and Wu (2020) measure housing affordability in China from both the macro and micro perspectives:
  - Homeowners: house price & repayment to income ratio
  - Renters: rent to income ratio
  - Panel data of city-level housing affordability: available online

- They find that the overall housing affordability in Chinese cities remained relatively stable or even improved between 2014 and 2018

- A few "superstar cities" are associated with serious housing affordability problem

Housing market related concerns in China

- Real HPs increase by 178% in 35 major cities in China b/w 2003-2018:
  - In US before Great Financial Crisis b/w 1991-2006: about 50%
  - In UK before Great Financial Crisis b/w 1991-2006: about 124%

- House price to income ratio is extremely high in Tier-1 and most Tier-2 cities in China

- Yet: High vacancy rates and over-supply of housing in some cities…

⇨ Is there a 'housing bubble' (irrational exuberance) in China? Or are there other explanations?

Discuss…

Possible hypotheses from the literature

- Hypothesis 1: Irrational or rational exuberance
- Hypothesis 2: Constrained land supply in conjunction with strong demand (like in the UK)
- Hypothesis 3: Gender ratio, marriage market, and cultural reasons
- Hypothesis 4: Urban amenities (educational and medical resources) associated with homeownership

(Or: Some combination of the four)

Hypothesis #1: Rational and irrational exuberance

- Irrational exuberance refers to investor enthusiasm that drives asset prices up to levels that aren't supported by fundamentals

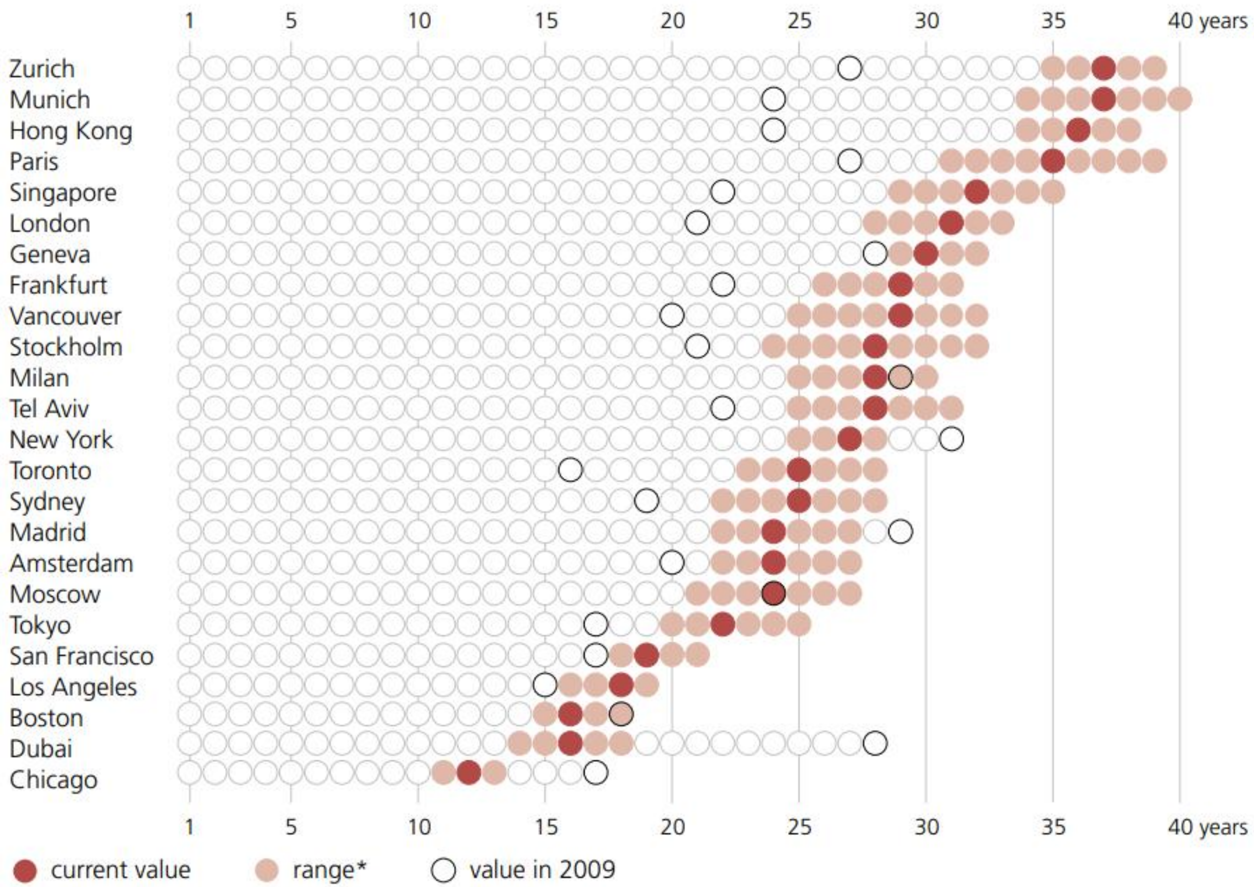
- Recall GY462…

$$House\ Price = \frac{Rent}{Cap\ Rate}$$

- Is housing price in China driven by fundamentals or irrational exuberance?

Price to rent ratios: international comparison

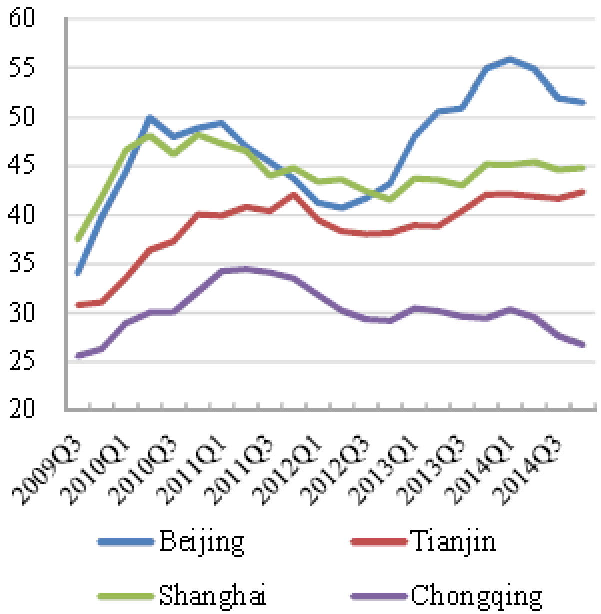
- According to UBS report (2019), Zurich has the highest price to rent ratio
- The ratios are around 31 in London and 27 in NY

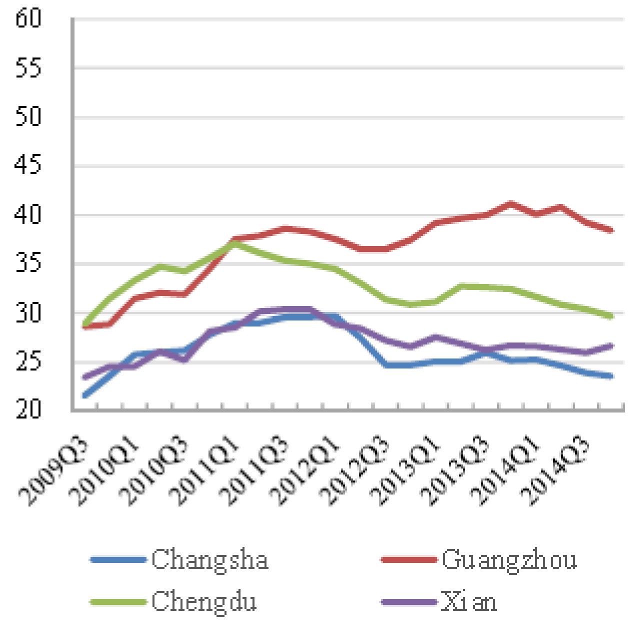


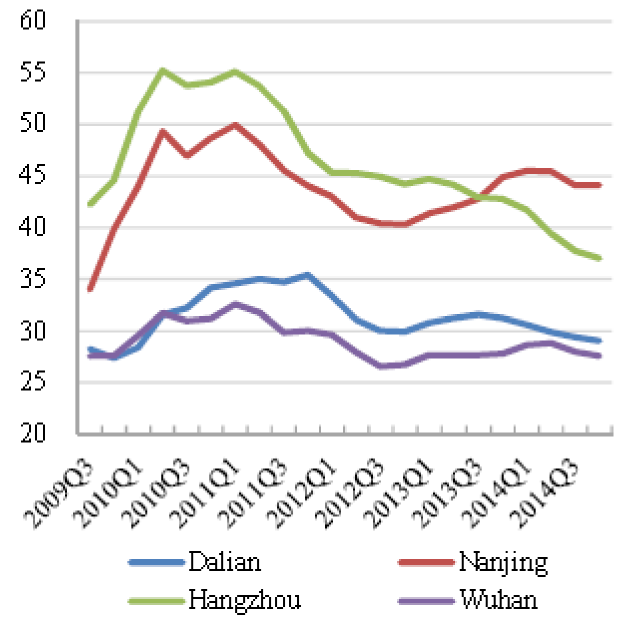
Source: UBS Global Real Estate Bubble Index (2019)

Price to rent ratios in several cities

- Price to rent ratios are extremely high in major cities in China, especially in Tier-1 cities
- The ratios are around 50 in Beijing and 45 in Shanghai (2014)







Source: Wu, Gyourko & Deng (2016)

Hypothesis #1: Rational and irrational exuberance

- Price to rent ratio is 50 in Beijing: **cap rate = 2%**
- Is it realistic? According to Gordon growth model:

$$House\ Price = \frac{Rent}{r - g}$$

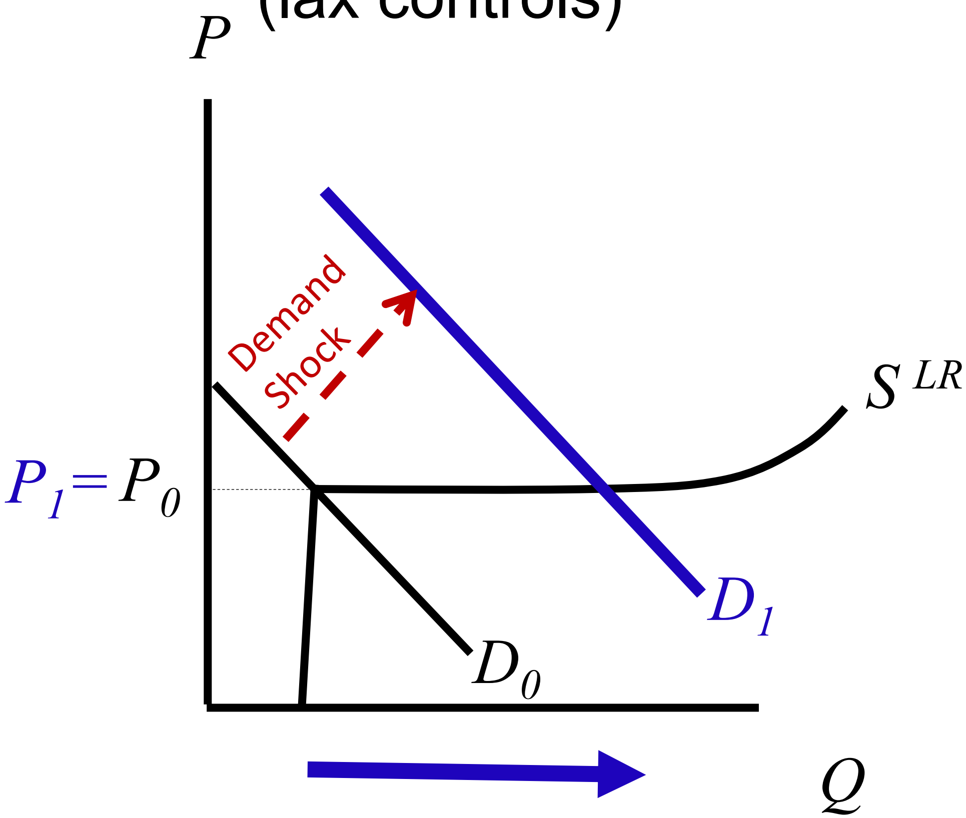
- Real GDP growth rate is above 6.5% in China between 1991 and 2018 – assuming rent growth rate is 4% going forward, the **implied r is 6%**
- Multifamily residential property's cap rate in the US is between 5 and 6% (CBRE cap rate survey, 2018) – assuming rent growth rate is 1%, the **implied r is 6-7%**

⇨ Irrational exuberance ('bubble') is not a likely explanation, if the expected rental growth rate is reasonable

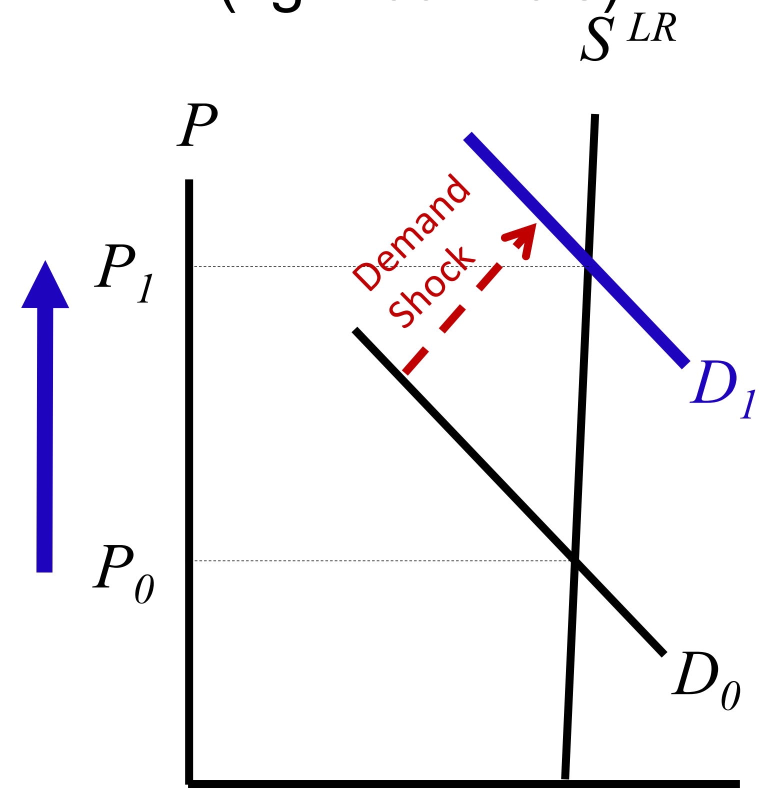
⇨ Recent economic performance: GDP growth rate is 3% in 2022, 5.4% in 2023, and 5% in 2024

Hypothesis #2: Constrained land supply in conjunction with strong demand

Housing market with elastic long-run supply upwards (lax controls)



Housing market with inelastic long-run supply upwards (tight controls)



Land acquisition in China

- Local governments acquire land parcels from farmers (via compensation) either at the urban fringe or near the city center, and then transfer land use rights to developers via land auctions

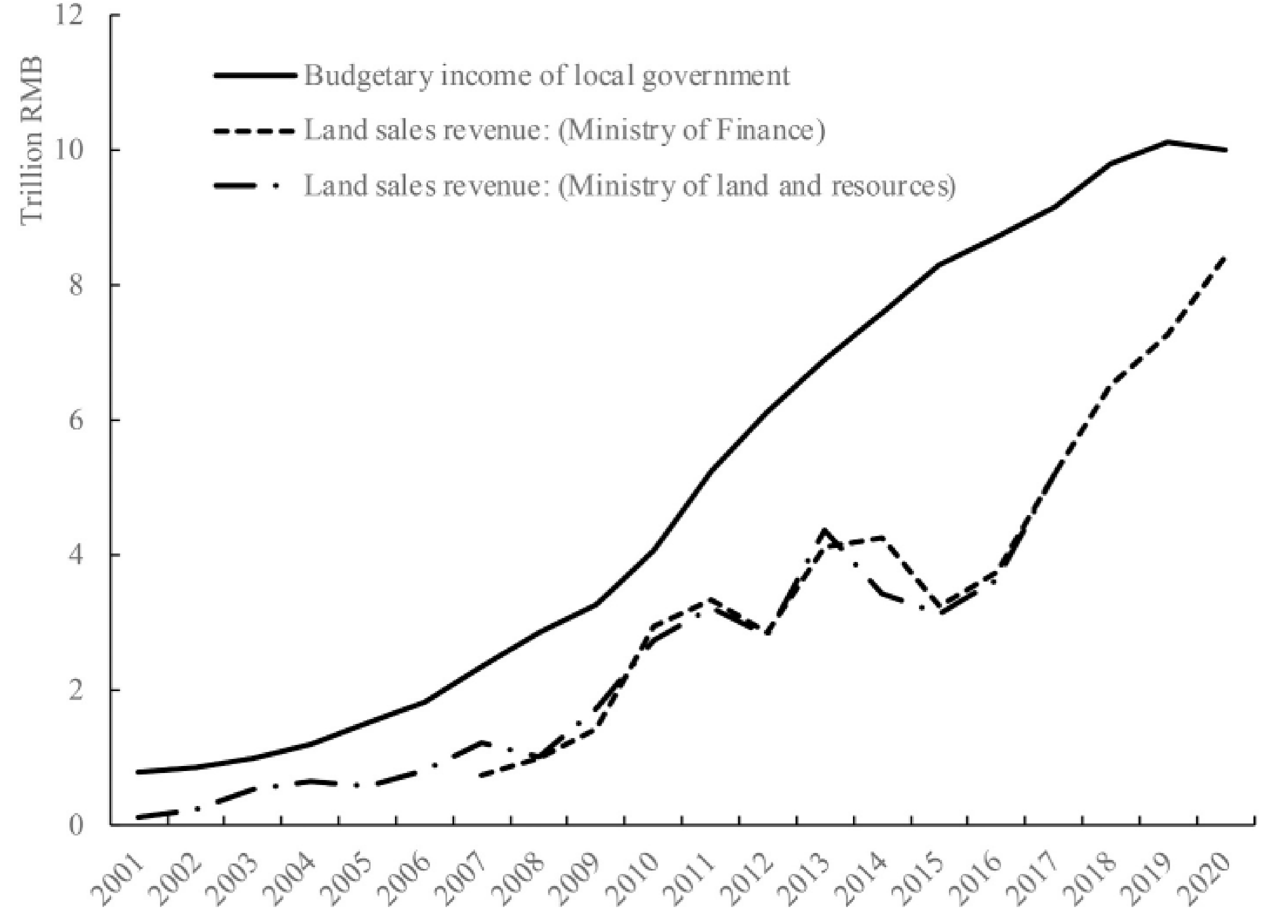
Agricultural lands at urban fringe

Urban village/informal housing in Guangzhou

- Abundant supply of informal housing helps accommodate migrant inflows and contain labor costs (Niu et al. 2020)

Local governments’ fiscal revenue and land sales

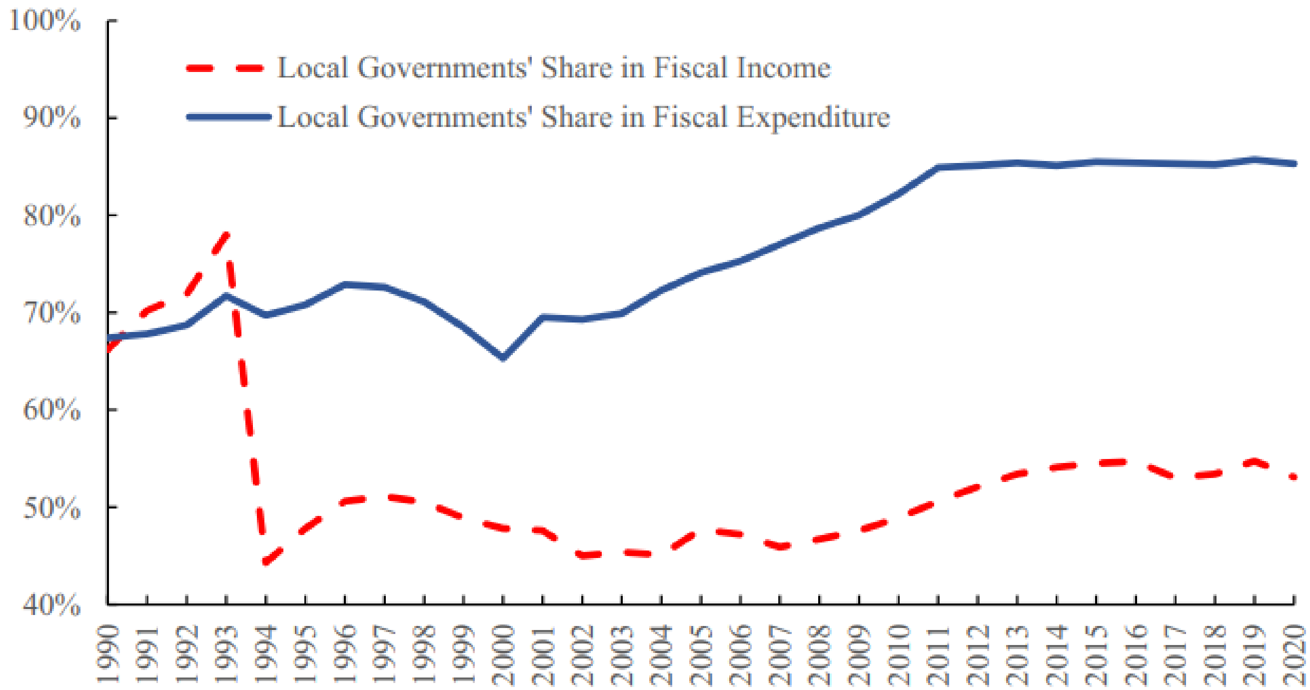
- Land sales revenue equals a high proportion of local governments’ budgetary revenue



Source: Gyourko et al. (2022)

Why land finance?

- Budgetary burden for local governments since the 1994 tax-share reform
- "Fiscal federalism, Chinese style": local governments have both a high degree of autonomy and a strong incentive to purse local economic development
- Local governments' high autonomy in determining land-related funding usage



Source: Gyourko et al. (2022)

Land use regulations in China

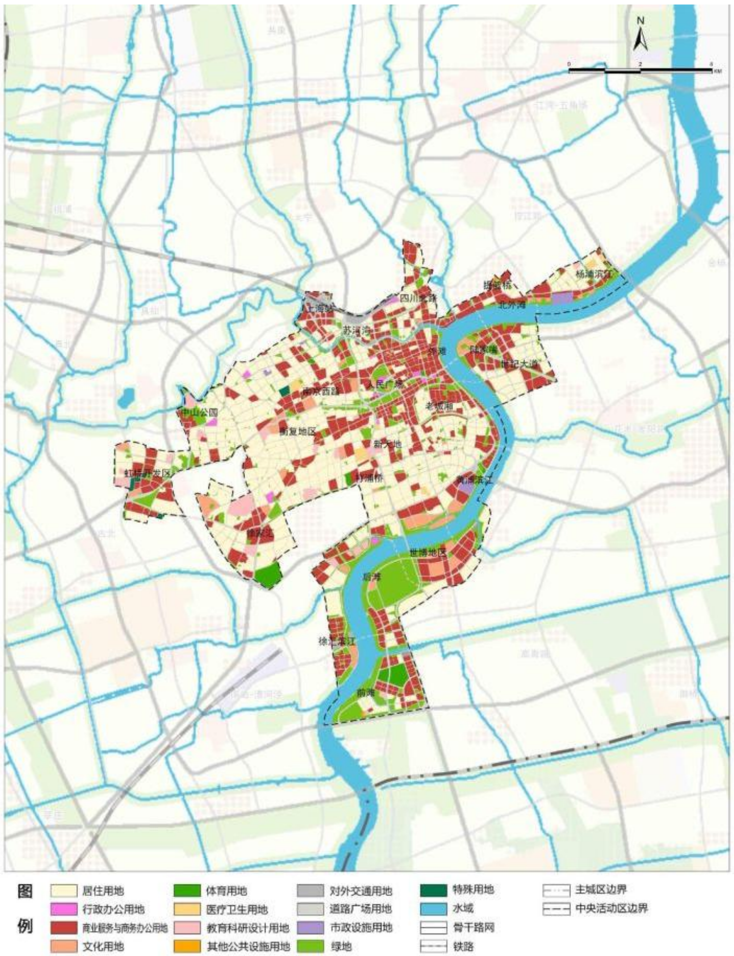
- The unique characteristic of urban planning in China is that many land use regulations are designed and implemented at the land parcel level

⇔ Zoning (e.g. in the US) or Development Control (UK)

- Land use regulations include:
  - ➢ Usage type, indicating whether the land is for residential, commercial, or industrial development
  - ➢ Construction density restriction (FAR limits), green coverage, and sometimes a separate explicit height limits, etc.

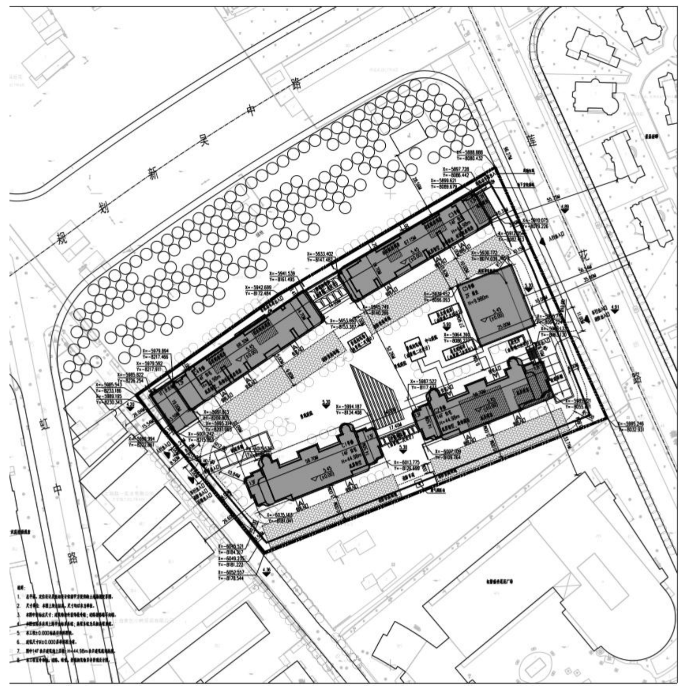
City-level and plot-level LURs

City-level plan for central Shanghai



Plot-level land use regulations

(Floor area ratio, building height limit, total construction area, etc)



经济技术指标

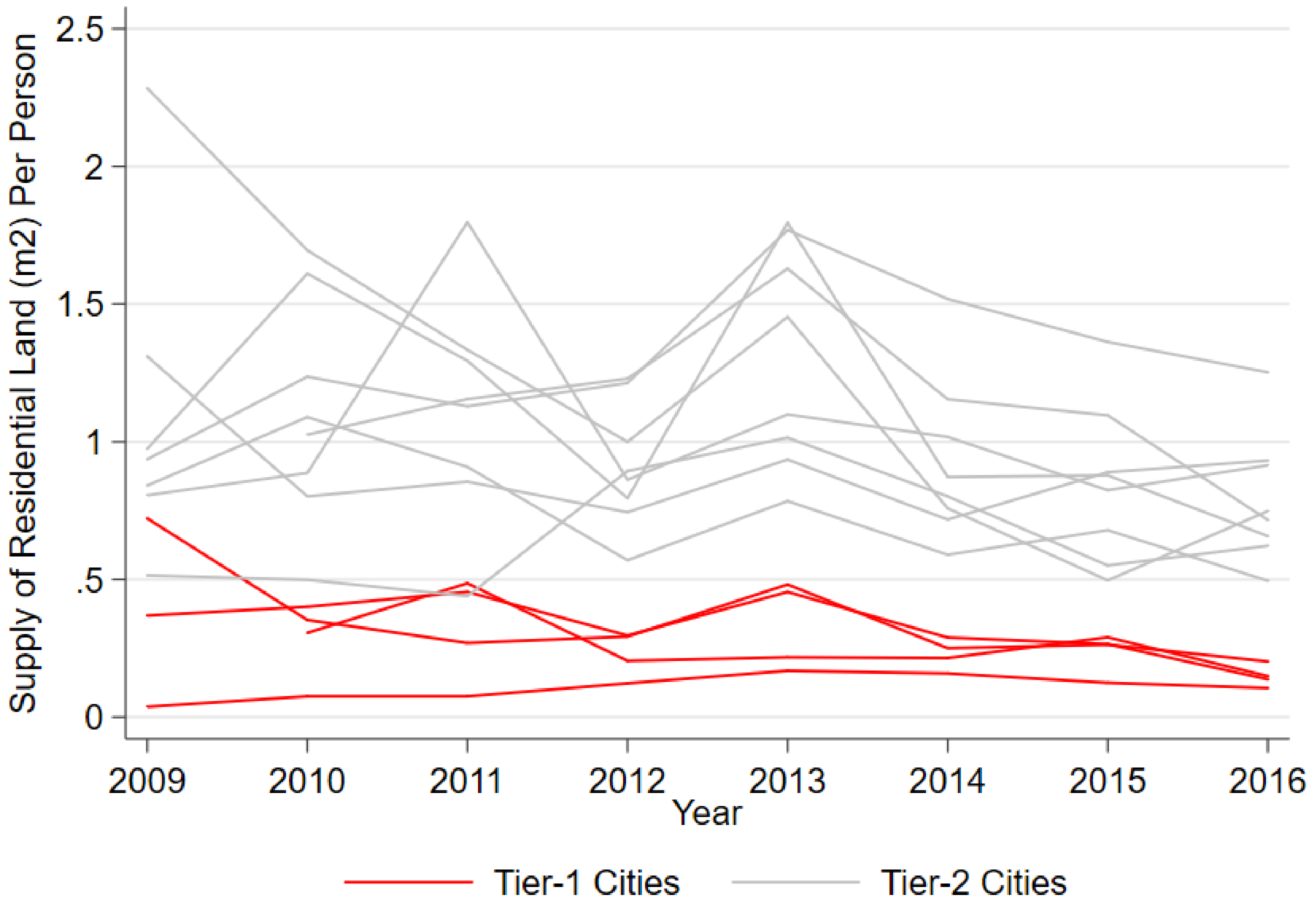
指标名称			数值	单位	备注
总用地面积			17220	平方米	
总建筑面积			80512.86	平方米	
其中	地上建筑面积		44912.86	平方米	
	其中	不计容建筑面积	2764.86	平方米	架空层，屋顶楼梯及设备机房面积
		计容建筑面积	42148.00	平方米	
	地下建筑面积		35600	平方米	
容积率			2.45		
绿地率			35.01%		

图例：
拟建建筑
用地红线
消防车道
地下室轮廓线
1F 建筑层数

Source: http://xxgk.shmh.gov.cn/mhxxgkweb/html/mh_xxgk/xxgk_ghj_ywxx_ghglgs/2019-03-21/Detail_62369.htm

Land supply in Tier-1 and Tier-2 cities

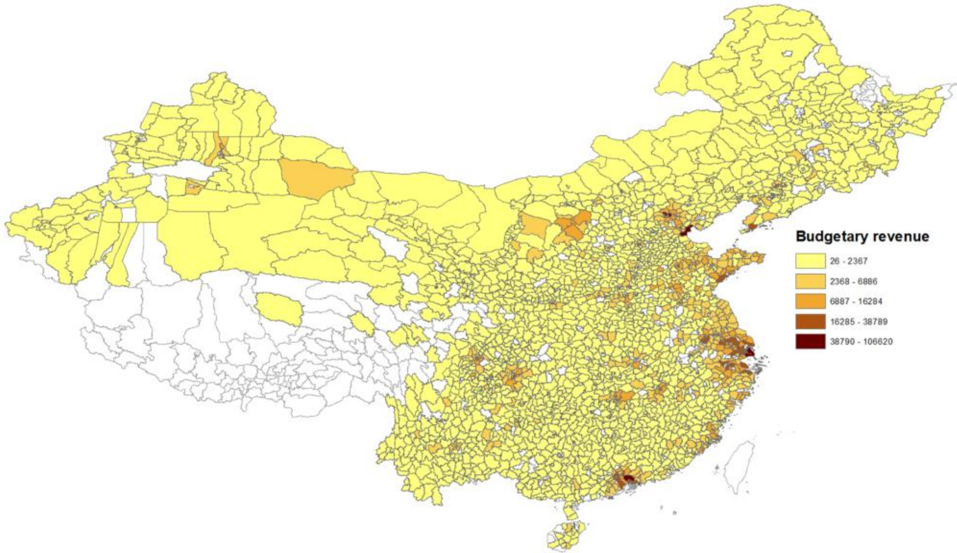
- Less land supply in Tier-1 cities compared with Tier-2 cities

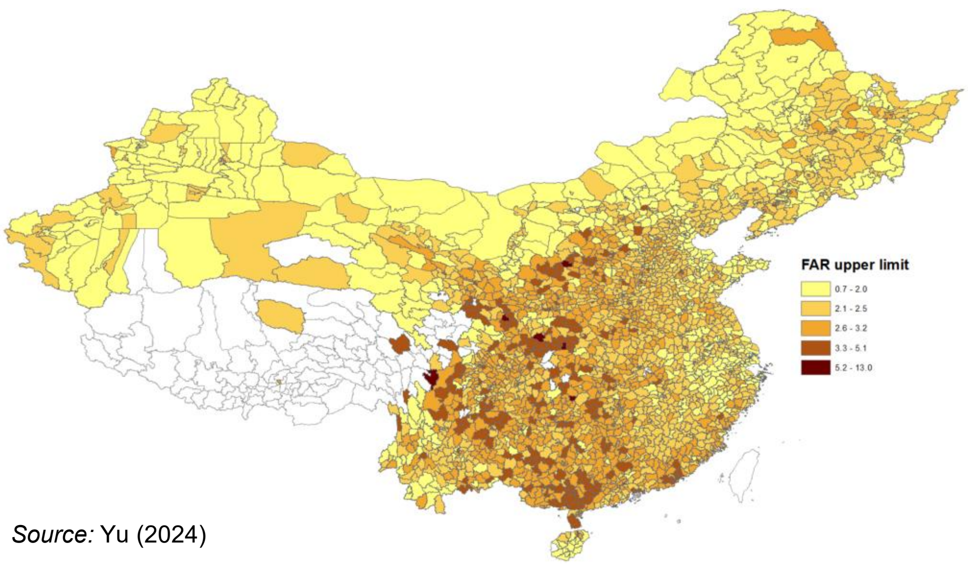


(despite strong demand in Tier 1 cities!)

Source: National Bureau of Statistics & self calculation

Land use regulation stringency in Chinese cities





Source: Yu (2024)

- Brueckner et al. (2017) develop a theory showing that the elasticity of land price with respect to floor area ratio (FAR), a building height indicator, is a measure of regulation's stringency. They find substantial variation in the stringency of FAR regulation across Chinese cities

- Yu (2024) finds that Shanghai (1.9) and Beijing (2.3) set lower FAR limits for residential land parcels relative to the national mean level (2.8)

Hypothesis #2: Constrained land supply

Possible **causes** for constrained land supply in Tier-1 cities:

Discuss…

Hypothesis #2: Constrained land supply

Possible **causes** for constrained land supply in Tier-1 cities:

1. Fewer available land plots: scarcity or tight regulation (e.g. 1.8 billion mu farmland red line policy)

2. Compensation costs for farmers are higher in Tier-1 cities, making it harder for local governments to collect and supply new land plots

3. Population cap policy in Beijing and Shanghai: By 2020, population in Beijing should be <23 million (21.7 million in 2017), and pop. in Shanghai should be <25 million (24.18 million in 2017) ⇨ Local governments release less land plots to market

4. Land use regulation, at least in Western countries, is the outcome of political economy game b/w owners of developed & owners of undeveloped land (Hilber and Robert-Nicoud, 2013)

   ⇨ There is more developed land in Tier-1 cities: Owners of developed land also politically more powerful in China!

Hypothesis #2: Constrained land supply

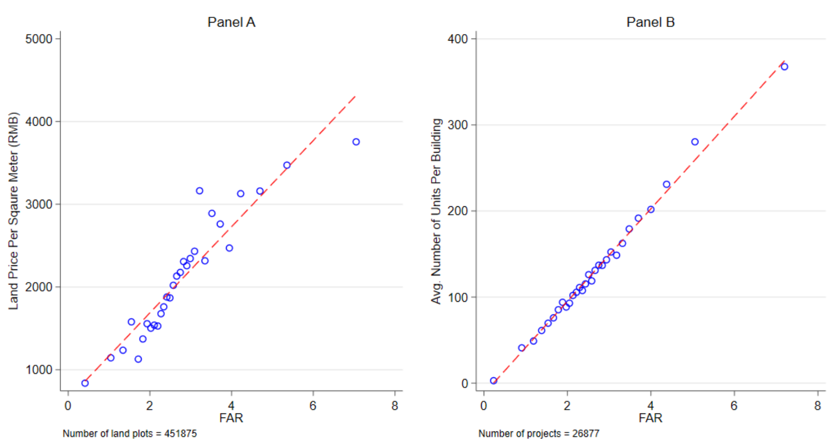
Possible **causes** for constrained land supply in Tier-1 cities:

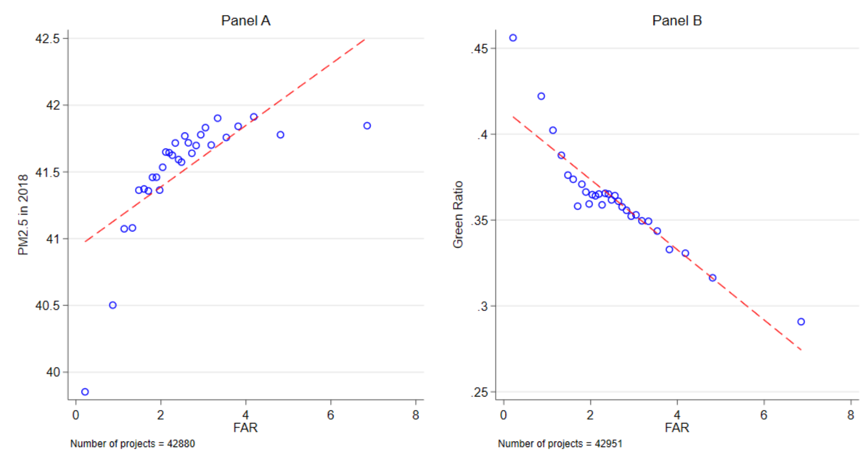
5. Hsu et al. (2017) find that corruption is highly correlated with an increase in land supply in China: less corruption in superstar cities? They also find that politicians' years in office also affect urban land supply: the impatience and anxiety in later years from not being promoted may contribute to an increase in land sales revenue
6. Wang et al. (2020) find that an increase in career incentive of city leader leads to outward spatial expansion

Hypothesis #2: Constrained land supply

Possible **causes** for constrained land supply in Tier-1 cities:

7. Yu (2024) proposes that local governments trade-off between the benefits and costs of high construction density: superstar cities are less financially reliant on land sale and put more weight on reducing negative externalities caused by high density

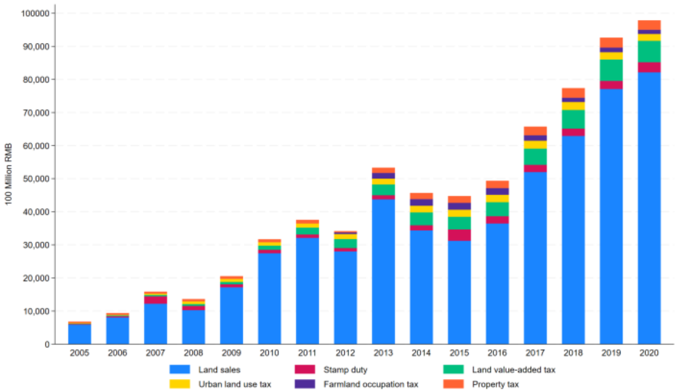




Hypothesis #2: Constrained land supply

Possible **causes** for constrained land supply in Tier-1 cities:

8. Different types of taxes and fees increase real estate development costs



Source: Yu (2026)

Table 1
Breakdown of land-related tax revenues.

Item	Volume (billion RMB)	Share in real estate–related taxes	Share in total tax revenue
1. Transaction Based			
(1.1) Deed tax	0.573	25.5%	7.6%
(1.2) Land value-added tax	0.564	25.1%	7.5%
(1.3) Farmland occupation tax	0.132	5.9%	1.7%
Subtotal	1.269	56.5%	16.8%
2. Stock Based			
(2.1) Property tax	0.289	12.9%	3.8%
(2.2) Urban land use tax	0.239	10.6%	3.2%
Subtotal	0.528	23.5%	7.0%
3. Developer Based			
(3.1) Developers' value-added tax	0.277	12.3%	3.7%
(3.2) Developers' business tax	0.004	0.2%	0.1%
(3.3) Developers' corporate income tax	0.169	7.5%	2.2%
Subtotal	0.451	20.1%	6.0%
Aggregated Real Estate–Related Taxes	2.248	100.0%	29.7%
Total Tax Revenue	7.562	–	100.0%

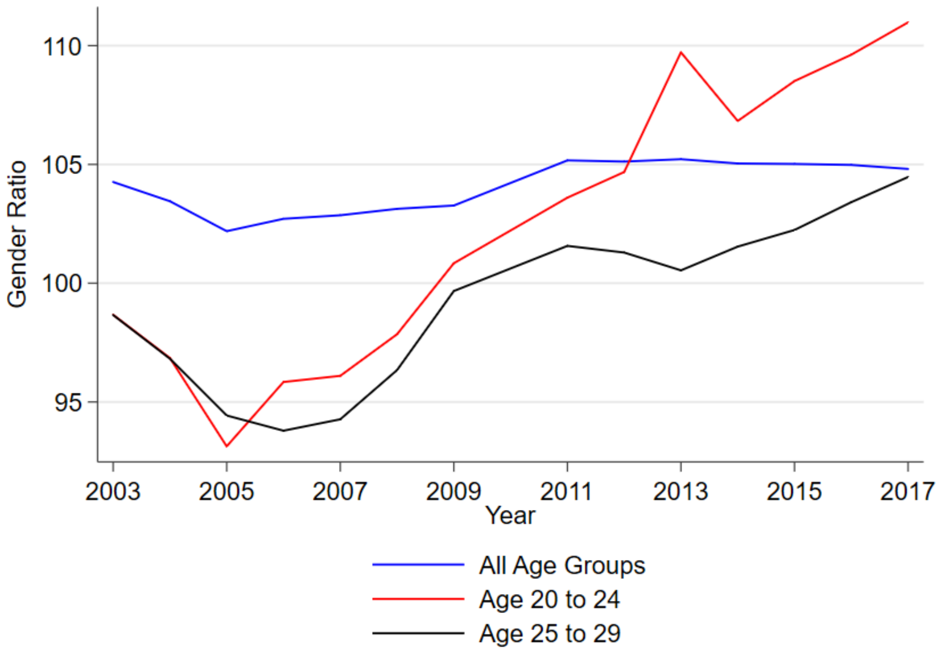
Note: Authors' calculations based on the data reported by the Ministry of Finance.

Source: Gyourko et al. (2025)

⇨ Supply constraints may explain to good extent why housing in Tier 1 cities may be so unaffordable…but more research needed!

Hypothesis #3: Gender ratio in China

- Male to female ratio (100=balanced) is increasing in China over past 15 years, especially for 20 to 29 yr. old
- Issue: To be successful on marriage market, men ought to possess a house ⇨ increases demand for housing



Source: National Bureau of Statistics & self calculation

Marriage market held at People's Park in Shanghai: Parents of unmarried adults flock to park and trade their children's information such as age, height, job, education **and property ownership (of the male)!**

Hypothesis #3: Gender ratio in China

- As the gender imbalance rises, Chinese parents with a son raise their savings in a competitive manner in order to improve their son's relative attractiveness for marriage (Wei and Zhang, 2011)
- A 10 basis point increase in the local male-to-female ratio is associated with a higher local cost of housing of about 4% (all else equal)

Table 11: Sex Ratios and Housing Values

LHS Variable = Per capita living space or average housing value (in log) at the county or city level in 2000

	County				City			
	Space	Space	Value	Value	Space	Space	Value	Value
Sex ratio for age cohort 10-19 in 2000	0.22**	-0.02	0.54**	0.37**	0.70**	0.37**	1.46**	0.74**
	(0.09)	(0.08)	(0.16)	(0.13)	(0.16)	(0.14)	(0.32)	(0.23)

Source: Wei & Zhang, 2011

Hypothesis #3: Cultural factors

- Saving rates are high for Chinese households, motivated by many factors from economics to culture (Chamon and Prasad 2010). Chinese savers have been pushed towards housing as a crucial form of investment:
  - ➢ Despite the staggering HP growth, the average real interest rate on Chinese bank deposits has been about zero over the past decade (Glaeser et al., 2017)
- Traditional agrarian culture in China cultivates a strong preference for owning properties:
  - ➢ Painter, Yang, and Yu (2003) find that home ownership is 18 pp higher among Chinese than among white households in the US

Hypothesis #3: "Six Wallets", Consumption and HPs

- Financial supports from parents increase demand for housing:
  - Fan Gang (president of China Development Institute) said that if a young couple has "six wallets", representing the financial support from the couple's two pairs of parents and four pairs of grandparents, the young couple could use that money to buy an apartment

⇨ May explain a part of the staggering increase in HPs

(But: again, more research needed!)

Hypothesis #4: Urban amenities

- In China, educational and medical resources are heavily concentrated in major cities, and the hukou system restricts access to many resources to local residents. High HPs could in part reflect a premium on urban amenities (Glaeser et al., 2017). In some cities, renters and homeowners don't have the same access to local public schools
- Chan et al. (2020) apply a boundary-discontinuity design in Shanghai and find that a superstar school causes housing prices in its neighborhood to increase by 14%, or about 430,000 RMB

Contents

Overview of housing policies

1. Purchase restriction policy
   - Minimum down payment requirements
   - Only households with local *hukou* (household registration) or who can prove they have worked locally for a consecutive number of years are eligible to purchase homes

2. Property tax
   - No nationwide property tax
   - Pilot programs in two cities since 2011: 0.6% or 0.4% in Shanghai; 0.5% to 1.2% in Chongqing. Both cities introduced exemption rules
   - Bai et al. (2015) find that property tax lowered the Shanghai average HP by 11%–15% but raised the Chongqing average HP by 10%–12% (driven by a potential spillover effect from high-end to low-end properties)
   - 2.73% more decrease in HPs for higher-taxed houses in Shanghai (Lyu, 2024)

3. Sales restriction policy
   - LGs prohibit re-sale of properties for a consecutive number of years (to limit speculative behavior)
   - LGs impose an upper-limit sale price for certain new builds/developments

4. Afford. housing scheme & shared ownership scheme

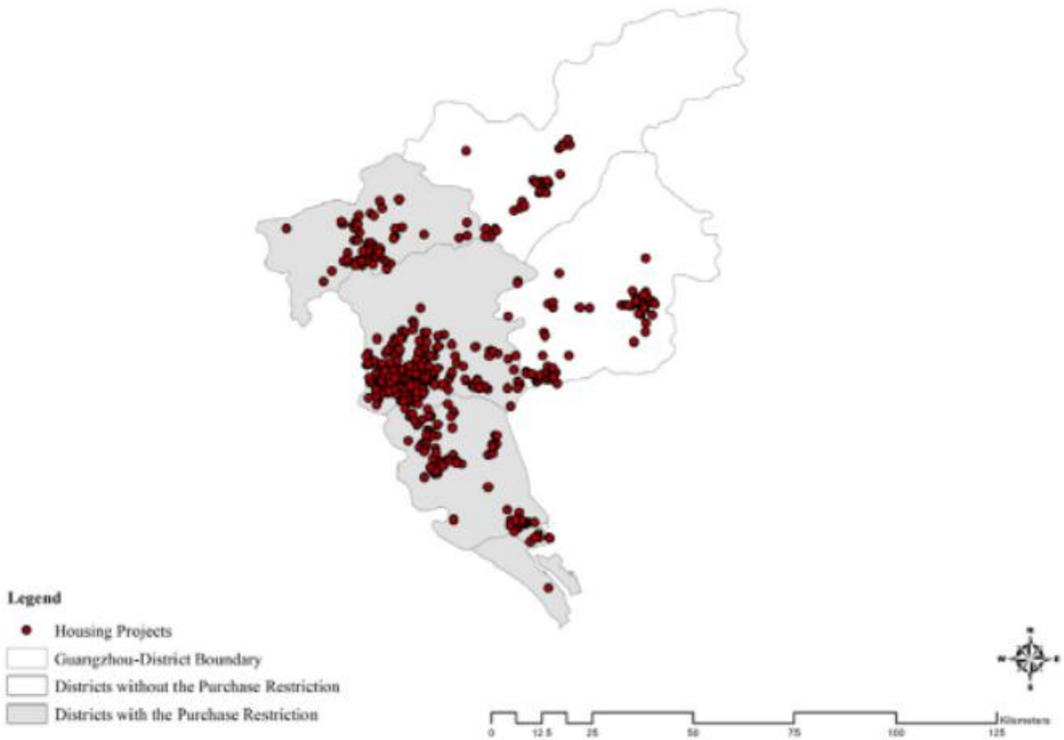
Policy evaluation

- Somerville *et al.* (2020) study the effect of home purchase restriction policy
  - ➢ Exploit variation in purchase restrictions within a city by district limits
- Data
  - ➢ Transaction prices & number of transactions (for newly constructed condominiums) come from China RE Information System (CREIS)
  - ➢ 2,014 development projects in Chengdu, Guangzhou, Hefei, and Qingdao
- Method: Diff-in-Diff

Policy evaluation

- The DiD estimation compares
  - HPs and number of transactions within a city by district
  - HPs close to the district border

Purchase Restriction by Districts and Housing Projects in Guangzhou



Policy evaluation

- Key findings
  1. In short run, restrictions on non-owner-occupant purchases significantly reduce purchase-activity levels by 40%
     - However, effects diminish over time
  2. There is no relative change in either the number of land auctions or prices paid for land
  3. There are no relative changes post-policy introduction in house prices b/w restricted & unrestricted areas (contradicting Du & Zhang 2015): downward sticky price response (loss aversion)
  4. While buyers were affected by the restrictions, developers behaved in a manner consistent with an expectation that the restrictions were temporary and that waiting to sell would be more profitable than dropping prices

Discussions

1. Purchase restriction policy
   - Could it address the affordability problem in the long term? Somerville et al. (2020) show that the restriction leads to fewer transactions but has an insignificant effect on house & land prices
   - Will young first-time buyers find it more difficult to afford homes due to the credit constraint? Will this policy discourage internal migration?

2. Property tax
   - The design of the property tax: second and above homes? high-end homes (spillover)? tax rates? redistributed? selective?
   - Will property taxes affect local housing markets adversely and influence local fiscal conditions?

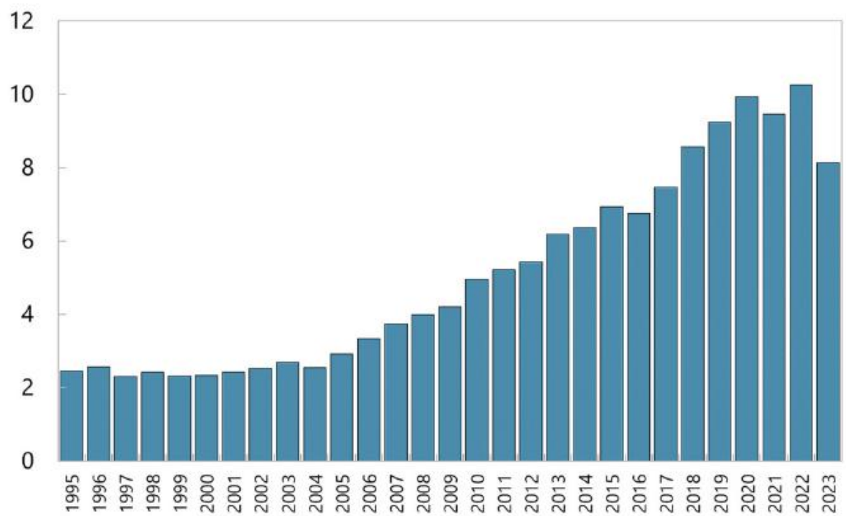
3. Affordable housing scheme
   - Who pays for the subsidy: local governments? developers?
   - How to design the allocation system?

Contents

Overbuilding?

- "Housing under construction over housing completed" indicates the number of years required to finish existing housing projects at the current construction rate
- A typical project requires 1-3 years for completion, and a high ratio can be expected in a rapidly growing market. However, ratios exceeding 10 might suggest a market in distress where developers struggle to finish projects because of a shortage of home purchasers and financing

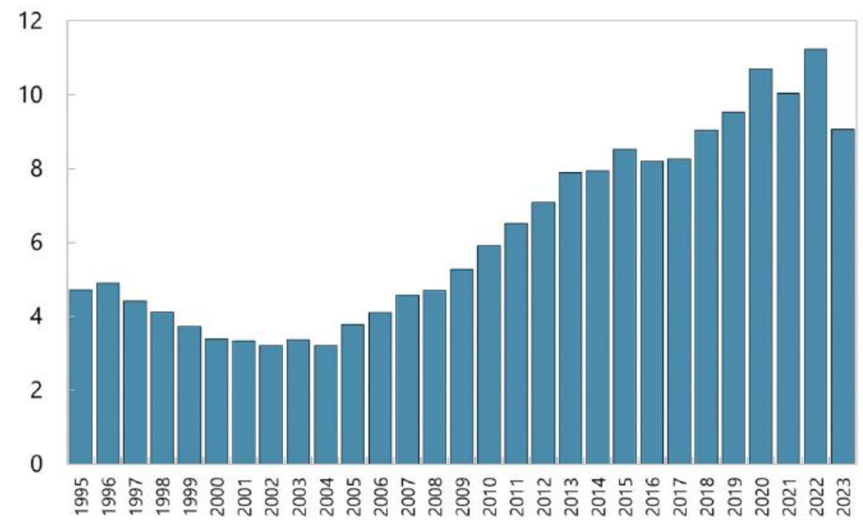
Housing Under Construction Over Housing Completed



Sources: NBS and author calculations

Fig. I.6. Housing Under Construction Over Housing Completed

CRE Under Construction Over CRE Completed



Sources: NBS and author calculations

Fig. I.7. CRE Under Construction Over CRE Completed

Source: Rogoff and Yang (2024)

"Ghost towns"

- between 2000 and 2011, newly constructed areas has increased by 76.4%, whereas the size of urban population has expanded by 50.5%... Motivated by land sale revenue, LGs prefer "constructing cities" (Wang et al., 2015)

An empty street in front of a vacant residential complex in Kangbashi, Ordos

Source: https://www.businessinsider.com/chinas-ghost-cities-in-2014-2014-6?r=US&IR=T

"Ghost towns"

- "There were roughly 63,000 people living in Kangbashi in 2013… While this district was originally planned for 300,000 people and most of the homes were still empty… in 2017, there are now 153,000 people living there, and housing prices have risen roughly 50% on average from the end of 2015. Of the 40,000 apartments that had been built in the new district since 2004, only 500 are still on the market" (Forbes, 2017): Planning and building ahead of urbanization?
- However, a joint study by Peking University and Baidu in 2015 still found high vacancy rates in parts of Kangbash

Sources:

https://qz.com/540571/baidu-found-chinas-ghost-cities-but-it-is-keeping-their-locations-mostly-a-secret/

https://www.forbes.com/sites/wadeshepard/2017/06/30/ordos-chinas-most-infamous-ex-ghost-city-continues-rising/?sh=743f5c7e6877#6aff89a46877/2017/06/30/

Vacancy rate

- Since 1985 no more *official* estimates available, but data is available from other sources

Source	Estimates	Year	Method
China Household Finance Survey by SWUFE	From 18.4% to 21.4%	2011, 2013, 2015 and 2017	Based on over 40,000 households in 364 districts or county-level cities in 29 provinces
China International Capital Corporation	From 17.7% to 18.3%	2010, 2013	Based on the aggregated data from the National Population Census in 2010
Wu, Gyourko & Deng (2016)	From 5.2% to 7%	2009, 2013	Based on Urban Household Survey by NBSC (9 provinces between 2002 and 2009; annual sample size: 50,000 households)
Glaeser et al. (2017)	20% in Tier 1 cities 13% in lower-tier cities	2012	Based on the Chinese Urban Household Survey (UHS)

⇨ Large discrepancies!

Vacancy rate — *cont.*

- The China Household Finance Survey in 2017 by SWUFE:
  - Over 40,000 households in 364 districts or county-level cities in 29 provinces
  - Home vacancy rates in China's urban areas were **18.4%**, **19.5%**, **20.6%** and **21.4%** in 2011, 2013, 2015 and 2017 respectively

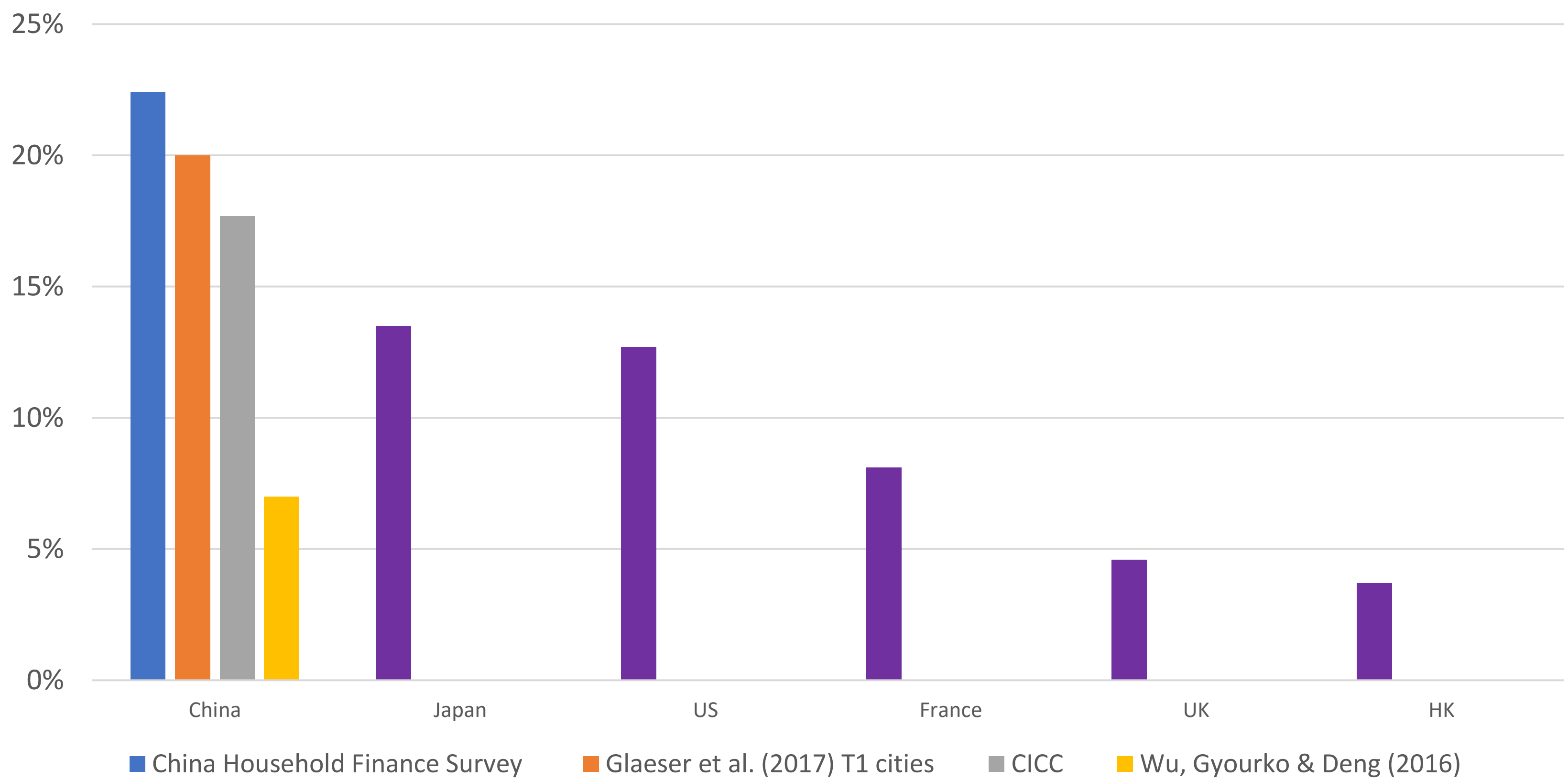
Bloomberg

Economics

A Fifth of China's Homes Are Empty. That's 50 Million Apartments

… and very high in international comparison…

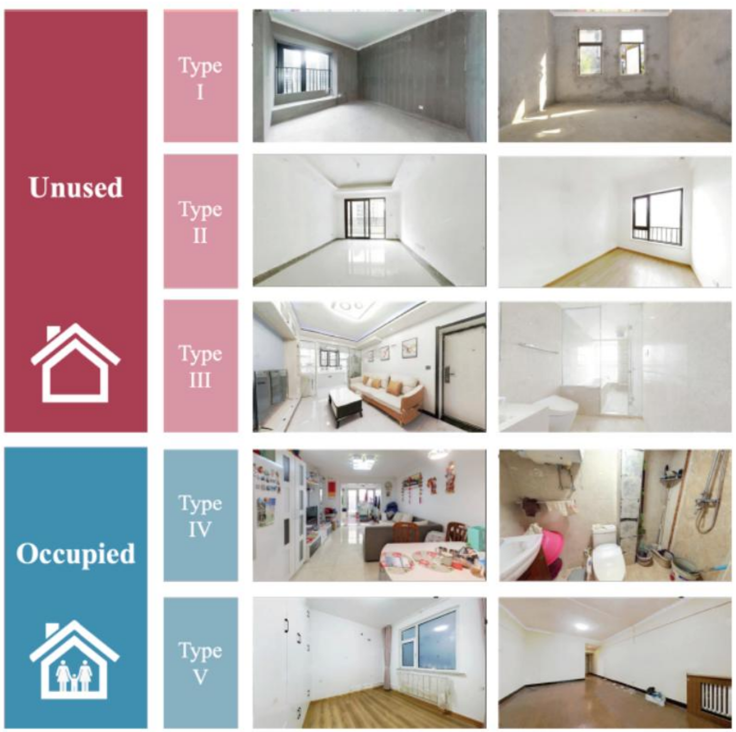
- Countries with inelastic long run supply of housing tend to have lower vacancy rates

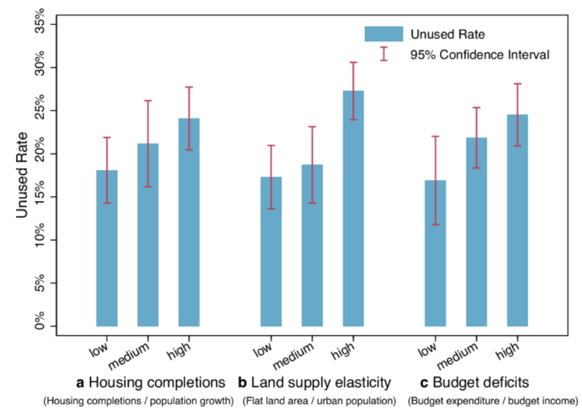


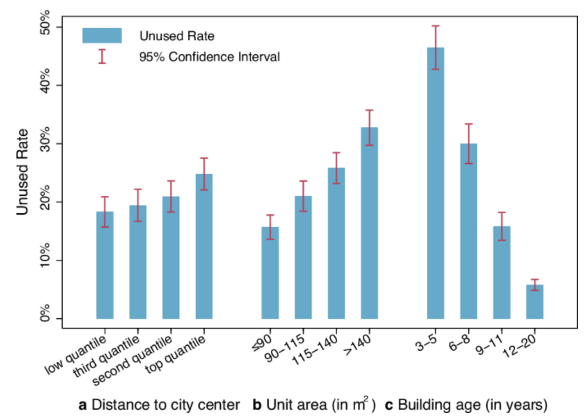
Note: Japan in 2013, UK in 2014, US, France and HK in 2017
Source: H.K's Rating and Valuation Department, Japan's statistics bureau, China Household Finance Survey, U.S. Housing Vacancy Survey
https://sdw.ecb.europa.eu/browse.do?node=9689360
https://www.bloomberg.com/news/articles/2018-11-08/a-fifth-of-china-s-homes-are-empty-that-s-50-million-apartments

Unused housing in China

- Zheng *et al.* (2025) define "unused housing" as dwelling units that have been built and sold for at least two years but never occupied. They find that by early 2021, 17.4% of the housing stock built in China (2000-2020) remained unused
- The unused rate is particularly high in cities more likely to be associated with the oversupply of housing, especially in the majority of third-tier cities



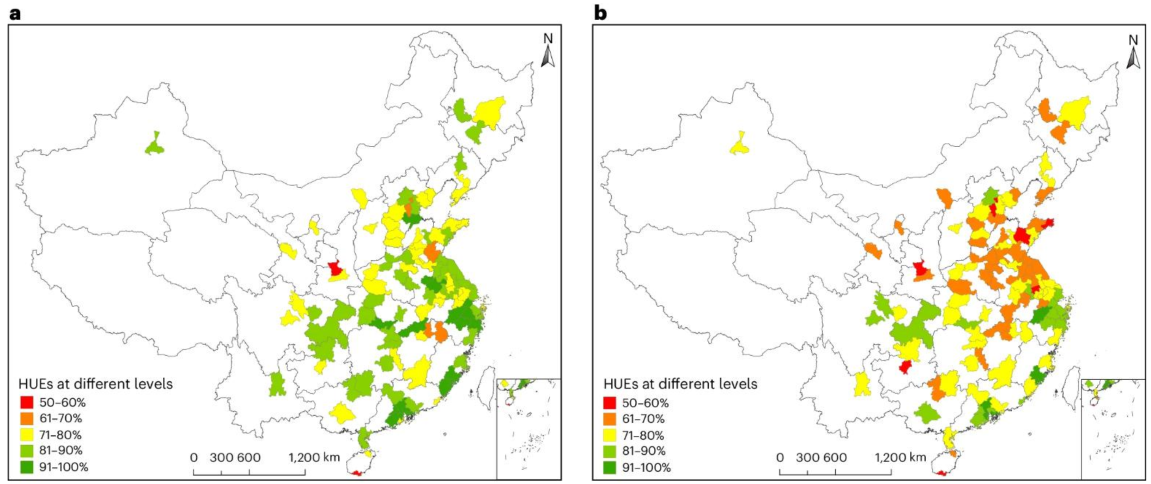




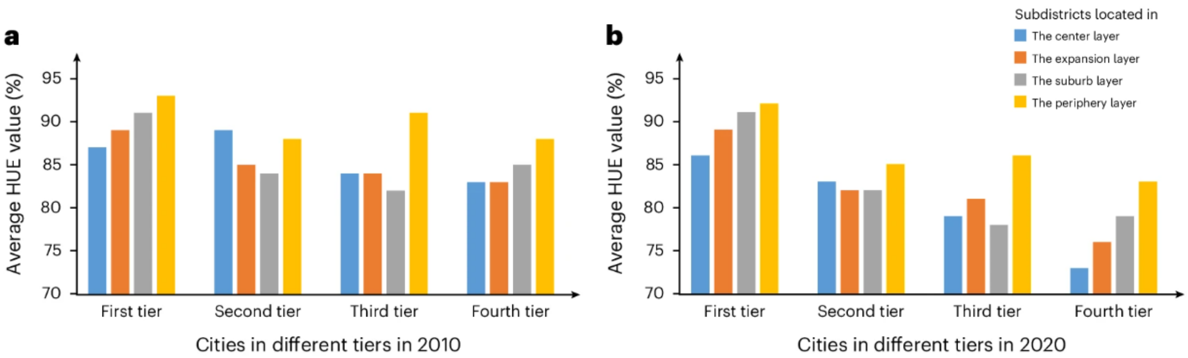
Source: Zheng *et al.* (2025)

Housing utilization efficiency

- Shi et al. (2025) propose the concept of housing utilization efficiency (HUE), measured as the ratio between the actual population of a subdistrict and the population capacity of all housing units physically suitable for occupancy in the subdistrict



a, Regional differences of HUEs across China in 2010. **b**, Regional differences of HUEs across China in 2020.



The authors find that the overall HUE in China's highly urbanized areas decreased from 84% in 2010 to 78% in 2020

Source: Shi *et al.* (2025)

Possible causes for high vacancy rate in China

⇨ How to explain the comparatively high vacancy rate in China?

Discuss…

Possible causes for high vacancy rate in China

⇨ How to explain the comparatively high vacancy rate in China?

Discuss…

1. High vacancy rate could be driven by speculative buyers: expect high house value appreciation in future (but still question: why do they not rent out?)
2. Perhaps the opportunity cost of not renting out is low: low rent to price ratio (at moment) and the capital gains of holding the property high
3. Perhaps cultural reasons: HHs in China prefer new-build (once occupied doesn't count as new anymore and can only sell at discount)

Possible causes for high vacancy rate in China

4. Owning empty property is not associated with any property taxes, which would make vacancies more attractive *(only trial program of property tax in Shanghai and Chongqing)*

5. Over-supply of land plots in some cities: local governors pursue for economic growth or corruption?

6. Rural-urban migration & properties back in hometowns?

➔ Still, more research needed!

Contents

Impact of fiscal stimulus program in 2008

- In 2008, China launched a massive fiscal stimulus program (4 trillion RMB ~ 12.5% of China's GDP) as an attempt to minimize the impact of the global financial crisis
- The stimulus package was invested in areas such as housing, infrastructure, transportation, health, education, environment, disaster rebuilding, etc.
- The program was seen as a success: China's economic growth recovered from 6% during 4Q 2008 and 1Q 2009 to over 8% in Q2 2009 and over 10% in Q3 2009
- However, the stimulus package has been criticized for causing a surge in local government debt in China after 2009

Impact of fiscal stimulus program in 2008

- Land and future land sale revenues serve as key collaterals for local governments to raise debt & finance infrastructure
- By 2013: Volume of outstanding local government debt reached 10.9 trillion RMB (~19.2% of China's GDP). 37% of local government's debt is collateralized by future land sale revenue (Liu and Xiong, 2018)
- Land revenue to repay the debt & LG's fiscal reliance on land revenue: one motivation for LG to implement restriction policies (instead of property tax) in China?

Real estate investment and GDP

- Real estate investment exceeds 10% of GDP in China in 2020 and 2021, higher than the US level before the Great Financial Crisis

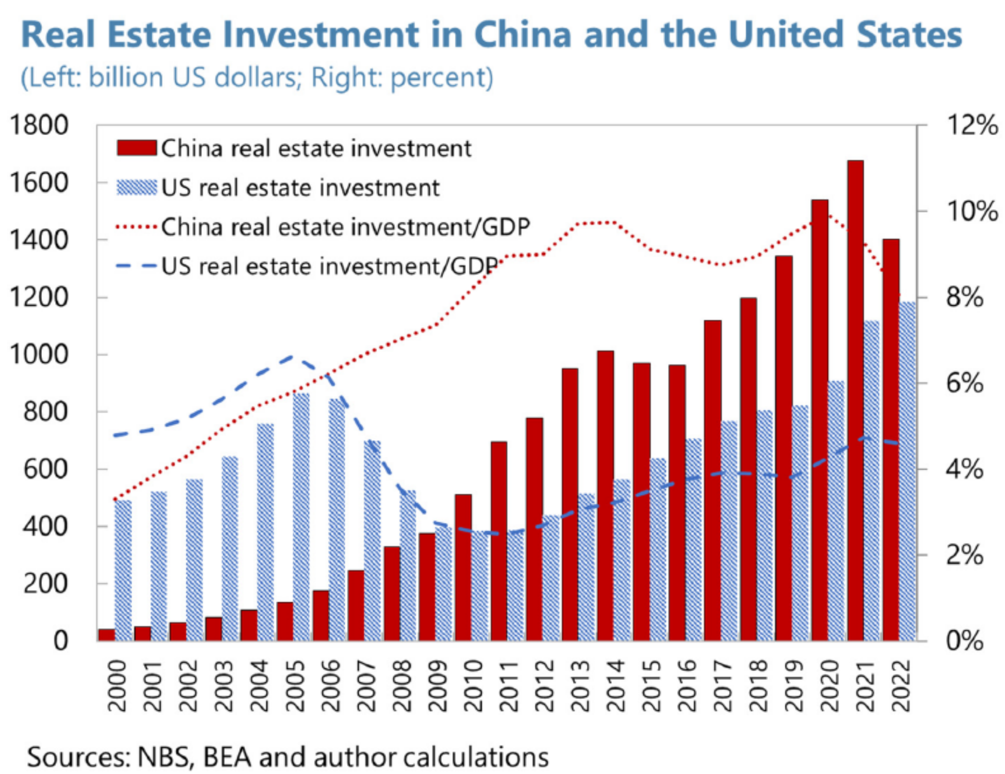


Fig. 2. Real Estate Investment in China and the United States.

Source: Rogoff and Yang (2024)

RE investment and local government debt

- Rogoff and Yang (2024) find that a 1 percentage increase in the RE investment ratio is associated with an approximately 3% increase in local governments' total debt scale

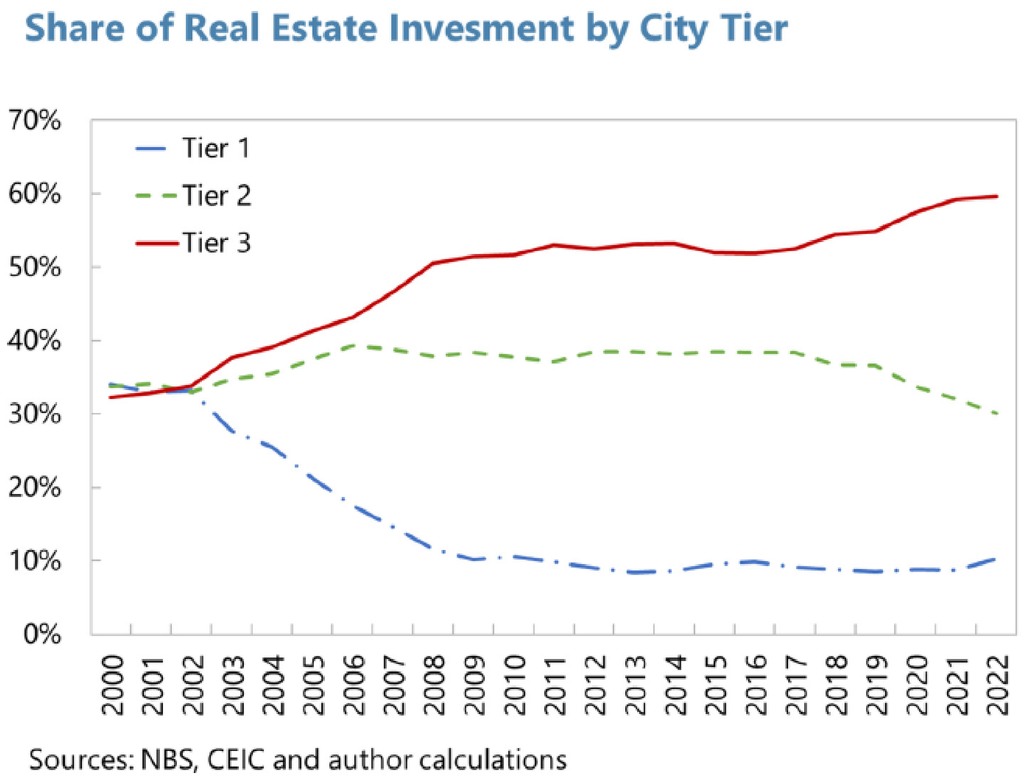
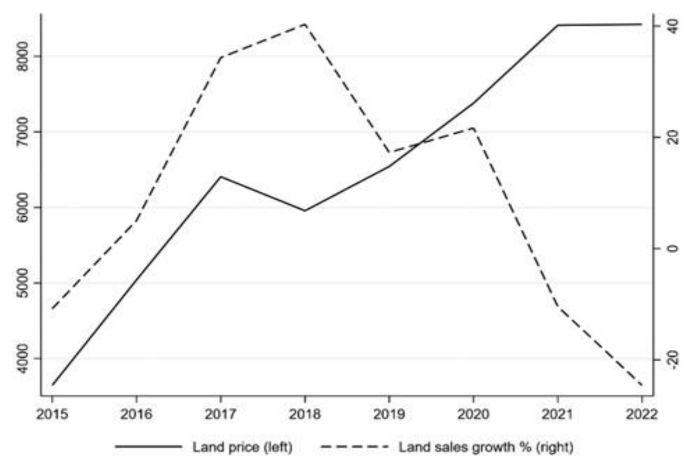


Fig. 5. Share of Real Estate Investment by City Tier.

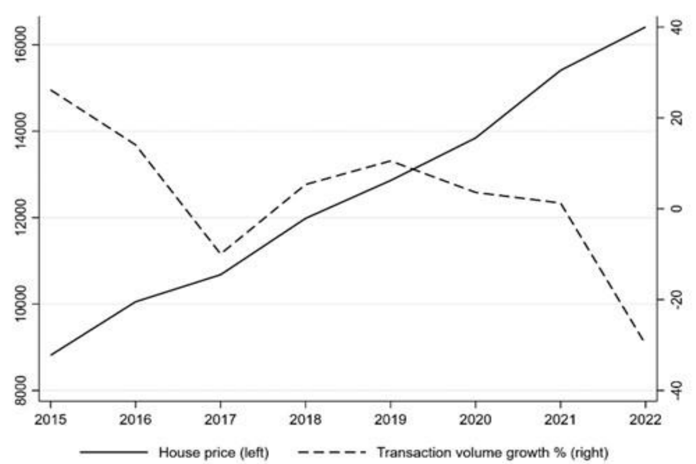
Source: Rogoff and Yang (2024)

Local governments and housing markets

A *Land price and transaction volume (area) growth rate*



B *New housing price and transaction volume (area) growth rate*



- Chang, Wang, and Xiong (2026) find that during the COVID pandemic (2020-2022), residential land and property prices surged even as transaction volumes plummeted

- The authors attribute this phenomenon to local governments' active price management through supply controls, land acquisitions by local government financing vehicles (LGFVs), and limits on new home sales permits

Source: Chang, Wang, and Xiong (2026)

Evergrande debt issue

- Evergrande is one of the largest RE developers in China
- In September 2021, the developer had 2 trillion RMB (310 billion USD) in liabilities
- In 2021, the company missed several debt payments and was downgraded by international ratings agencies:
  - ➢ Fitch downgraded Evergrande from B+ to CCC+
  - ➢ Moody‘s downgraded Evergrande’s rating from B2 to Caa1
  - ➢ S&P Global Ratings downgraded Evergrande from B− to CCC
  - ➢ Fitch downgraded Evergrande further from CCC+ to CC
- The company finally defaulted on an offshore bond at the beginning of December 2021
- On 29 January 2024, a Hong Kong-based court ordered Evergrade to liquidate due to a lack of a viable restructuring plan

Evergrande bond & stock prices

Source: https://www.bondsupermart.com/bsm/bond-factsheet/XS1627599654

Source: https://www.google.com/finance/quote/EGRNQ:OTCMKTS?sa=X&ved=2ahUKEwjk--TCtsuEAxWZ-AIHHcNXAIEQ3ecFegQIShAf&window=MAX

Evergrande debt issue: potential causes

- Expansion and diversification strategy:
  - Evergrande's land reserves alone were large enough to house 10 million people in 2020
  - Evergrande had pursued an aggressive expansion, including ventures in electric vehicles, theme parks, energy, etc.
- High financial leverage through wealth management products (WMP): total WMP liabilities stood at 40 billion RMB in September 2021
- Policies to control rising HPs:
  - The "three red lines": government regulates the leverage taken on by developers, limiting their borrowing based on the following metrics: debt-to-cash, debt-to-equity, and debt-to-assets
  - Sale price caps in some cities
- COVID, aging population, and the expected future growth rate (Gordon growth model), especially for the lower-Tier cities

A political economy perspective

- Li and Zhou (2005) employ turnover data of top provincial leaders and economic outcome data in China between 1979 and 1995
- Key findings:
  - The likelihood of promotion of provincial leaders increases with their economic performance
  - The likelihood of being 'sidelined' decreases with their economic performance
- Interpreted by authors as evidence that China's central government utilizes personnel control to induce desirable economic outcomes: politician tournament theory

Contents

Conclusions

- Real estate industry is crucial for China's economy. In 2017, housing sales totaled 13.37 trillion RMB, equivalent to 16.4% of China's GDP (Liu and Xiong, 2018)
- According to estimates from World Bank, China is the second largest economy in the world in terms of GDP (14.3 trillion USD ~ around 16% of World's GDP in 2019). The up and down of China's housing market also affect the global economy
- Government implemented different housing policies to address the affordability crisis: a comparative study of housing policies in the developed countries

Takeaways

- Many interesting research questions e.g. for REEF dissertations
  - Policy evaluation (whether/to what extent housing policies succeed in China?)
  - Why housing prices are so high in China given the high vacancy rate and price-to-income ratio?

- Data sources
  - National Bureau of Statistics, Department of land and resources
  - China City Statistical Yearbook, China Real Estate Statistics Yearbook
  - Wharton/Tsinghua Chinese Residential Land Price Indexes (CRLPI)
  - Brokers such as Lianjia, Fang.com, Anjuke
  - China Household Finance Survey by SWUFE (online application)
  - Orbis, Wind, Bloomberg
  - Housing affordability panel data (Li, Qin, and Wu 2020)

Reference

Bai, C., Li, Q., & Ouyang, M. (2014). Property taxes and home prices: A tale of two cities. *Journal of Econometrics*, *180*(1), 1-15.

Brueckner, J. K., Fu, S., Gu, Y., & Zhang, J. (2017). Measuring the stringency of land use regulation: The case of China's building height limits. *Review of Economics and Statistics*, *99*(4), 663-677.

Chamon, M. D., & Prasad, E. S. (2010). Why are saving rates of urban households in China rising?. *American Economic Journal: Macroeconomics*, *2*(1), 93-130.

Chan, J., Fang, X., Wang, Z., Zai, X., & Zhang, Q. (2020). Valuing primary schools in urban China. *Journal of Urban Economics*, *115*, 103183.

Chang, J., Wang, Y., & Xiong, W. (2026). Price and volume divergence in China's real estate markets: the role of local governments. *The Review of Financial Studies*, *39*(2), 343-386.

Chen, K., & Wen, Y. (2017). The great housing boom of China. *American Economic Journal: Macroeconomics*, *9*(2), 73-114.

Du, Z., & Zhang, L. (2015). Home-purchase restriction, property tax and housing price in China: A counterfactual analysis. *Journal of Econometrics*, *188*(2), 558-568.

Glaeser, E., Huang, W., Ma, Y., & Shleifer, A. (2017). A real estate boom with Chinese characteristics. *Journal of Economic Perspectives*, *31*(1), 93-116.

Gyourko, J., Shen, Y., Wu, J., & Zhang, R. (2022). Land finance in China: Analysis and review. *China Economic Review*, *76*, 101868.

Hilber, C. A., & Robert-Nicoud, F. (2013). On the origins of land use regulations: Theory and evidence from US metro areas. *Journal of Urban Economics*, *75*, 29-43.

Hsu, W. T., Li, X., Tang, Y., & Wu, J. (2017). Determinants of urban land supply in the people's republic of China: how do political factors matter?. *Asian Development Review*, *34*(2), 152-183.

Li, H., & Zhou, L. A. (2005). Political turnover and economic performance: the incentive role of personnel control in China. *Journal of public economics*, *89*(9-10), 1743-1762.

Li, K., Qin, Y., & Wu, J. (2020). Recent housing affordability in urban China: A comprehensive overview. *China Economic Review*, *59*, 101362.

Liu, C., & Xiong, W. (2018). China's Real Estate Market. NBER Working Paper No. 25297. *National Bureau of Economic Research*.

Lyu, X. (2024). Revisiting property tax capitalization. *Regional Science and Urban Economics*, *108*, 104039.

Reference — *cont.*

Niu, D., Sun, W., & Zheng, S. (2020). The Role of Informal Housing in Lowering China's Urbanization Costs. *Regional Science and Urban Economics*, 103638.

Painter, G., Yang, L., & Yu, Z. (2003). Heterogeneity in Asian American home-ownership: The impact of household endowments and immigrant status. *Urban Studies*, *40*(3), 505-530.

Rogoff, K. S., & Yang, Y. (2024). A tale of tier 3 cities. *Journal of International Economics*, *152*, 103989.

Somerville, T., Wang, L., & Yang, Y. (2019). Using purchase restrictions to cool housing markets: A within-market analysis. *Journal of Urban Economics*, 103189.

Shi, L., Leichtle, T., Huang, X., Wurm, M., & Taubenböck, H. (2025). The decreasing housing utilization efficiency in China's cities. *Nature Cities*, *2*(1), 70-80.

Song, Z., & Xiong, W. (2018). Risks in China's financial system. *Annual Review of Financial Economics*, *10*, 261-286.

United Nations (2025). *World Urbanization Prospects 2025: Summary of Results*. UN DESA/POP/2025/TR/ NO. 12. New York: United Nations.

Wang, X. R., Hui, E. C. M., Choguill, C., & Jia, S. H. (2015). The new urbanization policy in China: Which way forward?. *Habitat International*, *47*, 279-284.

Wang, Z., Zhang, Q., & Zhou, L. A. (2020). Career incentives of city leaders and urban spatial expansion in China. *Review of Economics and Statistics*, *102*(5), 897-911.

Wei, S. J., & Zhang, X. (2011). The competitive saving motive: Evidence from rising sex ratios and savings rates in China. *Journal of political Economy*, *119*(3), 511-564.

Xiong, W. (2023). *Derisking Real Estate in China's Hybrid Economy* (No. w31118). National Bureau of Economic Research.

Wu, J., Gyourko, J., & Deng, Y. (2012). Evaluating conditions in major Chinese housing markets. *Regional Science and Urban Economics*, *42*(3), 531-543.

Wu, J., Gyourko, J., & Deng, Y. (2016). Evaluating the risk of Chinese housing markets: What we know and what we need to know. *China Economic Review*, *39*, 91-114.

Yu, X. (2024). Low-rise buildings in big cities: Theory and evidence from China. *Real Estate Economics*, *52*(2), 366-400.

Yu, X. (2026). Regulatory tax, land, and housing markets: Theory and evidence from China. *Working paper.*
.

Zheng, H., Zhang, R., Yin, X., & Wu, J. (2025). Unused housing in urban China and its carbon emission impact. *Nature Communications*, *16*(1), 1985.